

To simplify further we will factor  $x^{-2}$  out of the square root. We need to be a little careful doing that. Take another look at examples 1.5.6 and 1.5.7 and the discussion between them before proceeding.

$$\begin{aligned}
 &= \frac{1}{\sqrt{x^{-2}(x^2-1)}} \cdot \frac{-1}{x^2} \\
 &= \frac{1}{|x^{-1}| \cdot \sqrt{x^2-1}} \cdot \frac{-1}{x^2} \quad \text{note that } x^2 \cdot |x^{-1}| = |x|. \\
 &= -\frac{1}{|x|\sqrt{x^2-1}}
 \end{aligned}$$

In the same way, we can find the derivative of the remaining inverse trig function. We just use its definition, a derivative we already know and the chain rule.

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{d}{dx} \arccos\left(\frac{1}{x}\right) = -\frac{1}{\sqrt{1-1/x^2}} \cdot \left(-\frac{1}{x^2}\right) = \frac{1}{|x|\sqrt{x^2-1}}$$

Example 2.12.6

By way of summary, we have

**Theorem 2.12.7.**

The derivatives of the inverse trigonometric functions are

$$\begin{aligned}
 \frac{d}{dx} \arcsin(x) &= \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \operatorname{arccsc}(x) &= -\frac{1}{|x|\sqrt{x^2-1}} \\
 \frac{d}{dx} \arccos(x) &= -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx} \operatorname{arcsec}(x) &= \frac{1}{|x|\sqrt{x^2-1}} \\
 \frac{d}{dx} \arctan(x) &= \frac{1}{1+x^2} & \frac{d}{dx} \operatorname{arccot}(x) &= -\frac{1}{1+x^2}
 \end{aligned}$$

## 2.13 ▲ The Mean Value Theorem

Consider the following situation. Two towns are separated by a 120km long stretch of road. The police in town *A* observe a car leaving at 1pm. Their colleagues in town *B* see the car arriving at 2pm. After a quick phone call between the two police stations, the driver is issued a fine for going 120km/h at some time between 1pm and 2pm. It is intuitively obvious<sup>48</sup> that, because his average velocity was 120km/h, the driver must have been going at least 120km/h at some point. From a knowledge of the average velocity of the car, we are able to deduce something about an instantaneous velocity<sup>49</sup>.

48 Unfortunately there are many obvious things that are decidedly false — for example “There are more rational numbers than integers.” or “Viking helmets had horns on them”.

49 Recall that speed and velocity are not the same.

Let us turn this around a little bit. Consider the premise of a 90s action film<sup>50</sup> — a bus must travel at a velocity of no less than  $80\text{km/h}$ . Being a bus, it is unable to go faster than, say,  $120\text{km/h}$ . The film runs for about 2 hours, and let's assume that there is about thirty minutes of non-action — so the bus' velocity is constrained between  $80$  and  $120\text{km/h}$  for a total of 1.5 hours.

It is again obvious that the bus must have travelled between  $80 \times 1.5 = 120$  and  $120 \times 1.5 = 180\text{km}$  during the film. This time, from a knowledge of the instantaneous rate of change of position — the derivative — throughout a 90 minute time interval, we are able to say something about the net change of position during the 90 minutes.

In both of these scenarios we are making use of a piece of mathematics called the Mean Value Theorem. It says that, under appropriate hypotheses, the average rate of change  $\frac{f(b)-f(a)}{b-a}$  of a function over an interval is achieved exactly by the instantaneous rate of change  $f'(c)$  of the function at some<sup>51</sup> (unknown) point  $a \leq c \leq b$ . We shall get to a precise statement in Theorem 2.13.4. We start working up to it by first considering the special case in which  $f(a) = f(b)$ .

## ► Rolle's Theorem

### Theorem 2.13.1 (Rolle's theorem).

Let  $a$  and  $b$  be real numbers with  $a < b$ . And let  $f$  be a function so that

- $f(x)$  is continuous on the closed interval  $a \leq x \leq b$ ,
- $f(x)$  is differentiable on the open interval  $a < x < b$ , and
- $f(a) = f(b)$

then there is a  $c$  strictly between  $a$  and  $b$ , i.e. obeying  $a < c < b$ , such that

$$f'(c) = 0.$$

Again, like the two scenarios above, this theorem says something intuitively obvious. Consider — if you throw a ball straight up into the air and then catch it, at some time in between the throw and the catch it must be stationary. Translating this into mathematical statements, let  $s(t)$  be the height of the ball above the ground in metres, and let  $t$  be time

- Velocity specifies the direction of motion as well as the rate of change. Objects moving along a straight line have velocities that are positive or negative numbers indicating which direction the object is moving along the line.
- Speed, on the other hand, is the distance travelled per unit time and is always a non-negative number — it is the absolute value of velocity.

50 The sequel won a Raspberry award for “Worst remake or sequel”.

51 There must be at least one such point — there could be more than one — but there cannot be zero.

from the moment the ball is thrown in seconds. Then we have

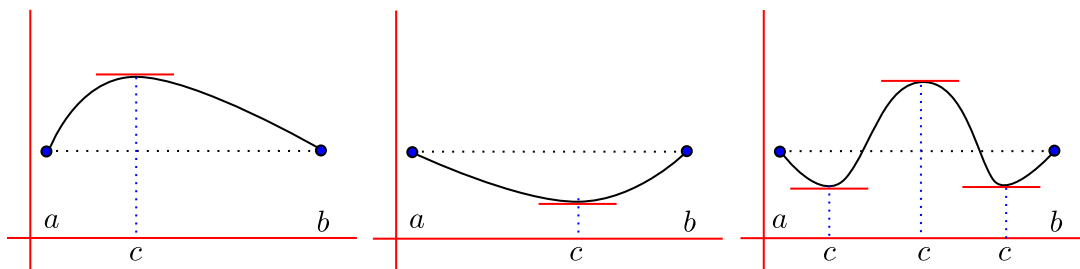
$$\begin{array}{ll} s(0) = 1 & \text{we release the ball at about hip-height} \\ s(4) = 1 & \text{we catch the ball 4s later at hip-height} \end{array}$$

Then we know there is some time in between — say at  $t = c$  — when the ball is stationary (in this case when the ball is at the top of its trajectory). I.e.

$$v(c) = s'(c) = 0.$$

Rolle's theorem guarantees that for any differentiable function that starts and ends at the same value, there will always be at least one point between the start and finish where the derivative is zero.

**Figure 2.13.1.**



There can, of course, also be multiple points at which the derivative is zero — but there must always be at least one. Notice, however, the theorem<sup>52</sup> does not tell us the value of  $c$ , just that such a  $c$  must exist.

### Example 2.13.2

We can use Rolle's theorem to show that the function

$$f(x) = \sin(x) - \cos(x)$$

has a point  $c$  between 0 and  $\frac{3\pi}{2}$  so that  $f'(c) = 0$ .

To apply Rolle's theorem we first have to show the function satisfies the conditions of the theorem on the interval  $[0, \frac{3\pi}{2}]$ .

- Since  $f$  is the sum of sine and cosine it is continuous on the interval and also differentiable on the interval.
- Further, since

$$\begin{aligned} f(0) &= \sin 0 - \cos 0 = 0 - 1 = -1 \\ f\left(\frac{3\pi}{2}\right) &= \sin \frac{3\pi}{2} - \cos \frac{3\pi}{2} = -1 - 0 = -1 \end{aligned}$$

we can now apply Rolle's theorem.

<sup>52</sup> Notice this is very similar to the intermediate value theorem (see Theorem 1.6.12)

- Rolle's theorem implies that there must be a point  $c \in (0, 3\pi/2)$  so that  $f'(c) = 0$ .

While Rolle's theorem doesn't tell us the value of  $c$ , this example is sufficiently simple that we can find it directly.

$$f'(x) = \cos x + \sin x$$

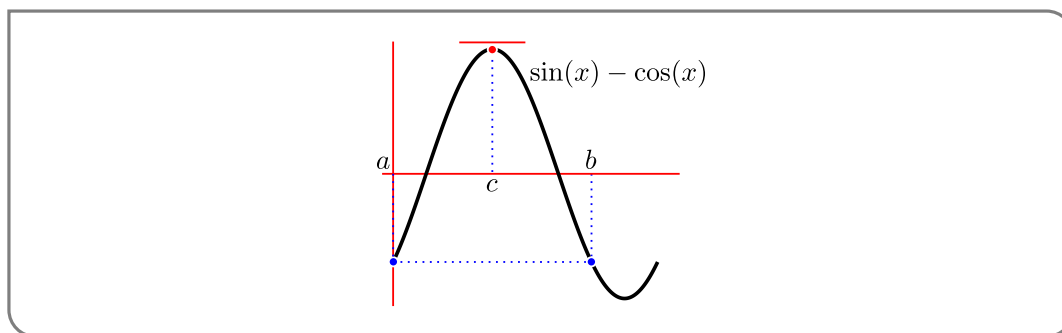
$$f'(c) = \cos c + \sin c = 0$$

$$\sin c = -\cos c$$

$$\tan c = -1$$

rearrange  
and divide by  $\cos c$

Hence  $c = \frac{3\pi}{4}$ . We have sketched the function and the relevant points below.



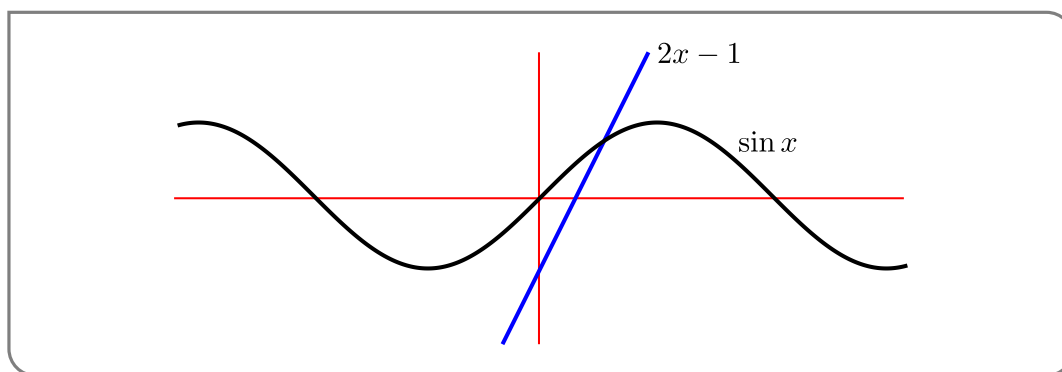
Example 2.13.2

A more substantial application of Rolle's theorem (in conjunction with the intermediate value theorem — Theorem 1.6.12) is to show that a function does not have multiple zeros in an interval:

Example 2.13.3

Show that the equation  $2x - 1 = \sin(x)$  has exactly 1 solution.

- Start with a rough sketch of each side of the equation



This seems like it should be true.

- Notice that the problem we are trying to solve is equivalent to showing that the function

$$f(x) = 2x - 1 - \sin(x)$$

has only a single zero.

- Since  $f(x)$  is the sum of a polynomial and a sine function, it is continuous and differentiable everywhere. Thus we can apply both the IVT and Rolle's theorem.
- Notice that  $f(0) = -1$  and  $f(2) = 4 - 1 - \sin(2) = 3 - \sin(2) \geq 2$ , since  $-1 \leq \sin(2) \leq 1$ . Thus by the IVT we know there is at least one number  $c$  between 0 and 2 so that  $f(c) = 0$ .
- But our job is only half done — this shows that there is at least one zero, but it does not tell us there is no more than one. We have more work to do, and Rolle's theorem is the tool we need.
- Consider what would happen if  $f(x)$  is zero in 2 places — that is, there are numbers  $a, b$  so that  $f(a) = f(b) = 0$ .
  - Since  $f(x)$  is differentiable everywhere and  $f(a) = f(b) = 0$ , we can apply Rolle's theorem.
  - Hence we know there is a point  $c$  between  $a$  and  $b$  so that  $f'(c) = 0$ .
  - But let us examine  $f'(x)$ :

$$f'(x) = 2 - \cos x$$

Since  $-1 \leq \cos x \leq 1$ , we must have that  $f'(x) \geq 1$ .

- But this contradicts Rolle's theorem which tells us there must be a point at which the derivative is zero.

Thus the function cannot be zero at two different places — otherwise we'd have a contradiction.

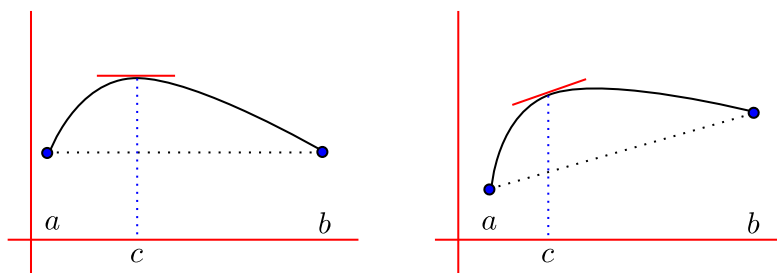
We can actually nail down the value of  $c$  using the bisection approach we used in example 1.6.15. If we do this carefully we find that  $c \approx 0.887862 \dots$

Example 2.13.3

## ►► Back to the MVT

Rolle's theorem can be generalised in a straight-forward way; given a differentiable function  $f(x)$  we can still say something about  $\frac{df}{dx}$ , even if  $f(a) \neq f(b)$ . Consider the following sketch:

Figure 2.13.2.



All we have done is tilt the picture so that  $f(a) < f(b)$ . Now we can no longer guarantee that there will be a point on the graph where the tangent line is horizontal, but there will be a point where the tangent line is parallel to the secant joining  $(a, f(a))$  to  $(b, f(b))$ .

To state this in terms of our first scenario back at the beginning of this section, suppose that you are driving along the  $x$ -axis. At time  $t = a$  you are at  $x = f(a)$  and at time  $t = b$  you are at  $x = f(b)$ . For simplicity, let's suppose that  $b > a$  and  $f(b) \geq f(a)$ , just like in the above sketch. Then during the time interval in question you travelled a net distance of  $f(b) - f(a)$ . It took you  $b - a$  units of time to travel that distance, so your average velocity was  $\frac{f(b) - f(a)}{b - a}$ . You may very well have been going faster than  $\frac{f(b) - f(a)}{b - a}$  part of the time and slower than  $\frac{f(b) - f(a)}{b - a}$  part of the time. But it is reasonable to guess that at some time between  $t = a$  and  $t = b$  your instantaneous velocity was exactly  $\frac{f(b) - f(a)}{b - a}$ . The mean value theorem says that, under reasonable assumptions about  $f$ , this is indeed the case.

**Theorem 2.13.4** (The mean value theorem).

Let  $a$  and  $b$  be real numbers with  $a < b$ . And let  $f(x)$  be a function so that

- $f(x)$  is continuous on the closed interval  $a \leq x \leq b$ , and
- $f(x)$  is differentiable on the open interval  $a < x < b$

then there is a  $c \in (a, b)$ , such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

which we can also express as

$$f(b) = f(a) + f'(c)(b - a).$$

Let us start to explore the mean value theorem — which is very frequently known as the MVT. A simple example to start:

**Example 2.13.5**

Consider the polynomial  $f(x) = 3x^2 - 4x + 2$  on  $[-1, 1]$ .

- Since  $f$  is a polynomial it is continuous on the interval and also differentiable on the interval. Hence we can apply the MVT.
- The MVT tells us that there is a point  $c \in (-1, 1)$  so that

$$f'(c) = \frac{f(1) - f(-1)}{1 - (-1)} = \frac{1 - 9}{2} = -4$$

This example is sufficiently simple that we can find the point  $c$  and the corresponding tangent line:

- The derivative is

$$f'(x) = 6x - 4$$

- So we need to solve  $f'(c) = -4$ :

$$6c - 4 = -4$$

which tells us that  $c = 0$ .

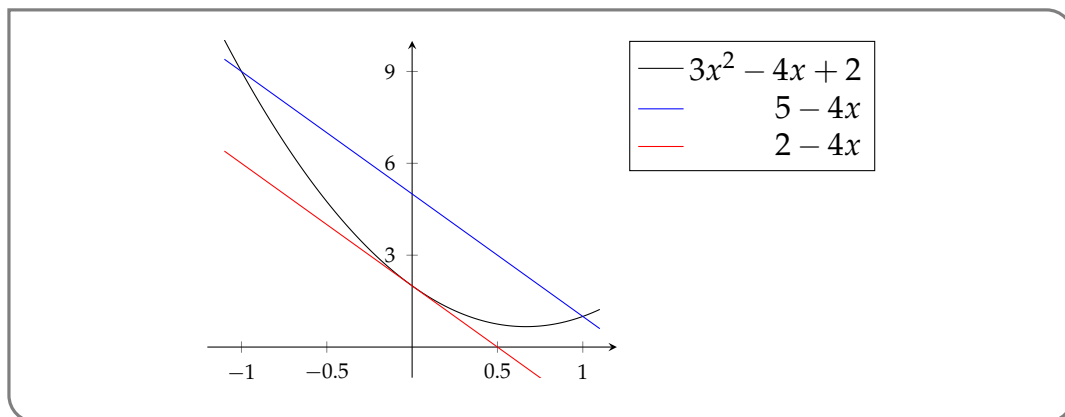
- The tangent line has slope  $-4$  and passes through  $(0, f(0)) = (0, 2)$ , and so is given by

$$y = -4x + 2$$

- The secant line joining  $(-1, f(-1)) = (-1, 9)$  to  $(1, f(1)) = (1, 1)$  is just

$$y = 5 - 4x$$

- Here is a sketch of the curve and the two lines:



Example 2.13.5

Example 2.13.6

We can return to our initial car-motivated examples. Say you are driving along a straight road in a car that can go at most  $80\text{km/h}$ . How far can you go in 2 hours? — the answer is easy, but we can also solve this using MVT.

- Let  $s(t)$  be the position of the car in  $km$  at time  $t$  measured in hours.
- Then  $s(0) = 0$  and  $s(2) = q$ , where  $q$  is the quantity that we need to bound.
- We are told that  $|s'(t)| \leq 80$ , or equivalently

$$-80 \leq s'(t) \leq 80$$

- By the MVT there is some  $c$  between 0 and 2 so that

$$s'(c) = \frac{q - 0}{2} = \frac{q}{2}$$

- Now since  $-80 \leq s'(c) \leq 80$  we must have  $-80 \leq q/2 \leq 80$  and hence  $-160 \leq q = s(2) \leq 160$ .

Example 2.13.6

More generally if we have some information about the derivative, then we can use the MVT to leverage this information to tell us something about the function.

Example 2.13.7

Let  $f(x)$  be a differentiable function so that

$$f(1) = 10 \quad \text{and} \quad -1 \leq f'(x) \leq 2 \text{ everywhere}$$

Obtain upper and lower bounds on  $f(5)$ .

Okay — what do we do?

- Since  $f(x)$  is differentiable we can use the MVT.
- Say  $f(5) = q$ , then the MVT tells us that there is some  $c$  between 1 and 5 such that

$$f'(c) = \frac{q - 10}{5 - 1} = \frac{q - 10}{4}$$

- But we know that  $-1 \leq f'(c) \leq 2$ , so

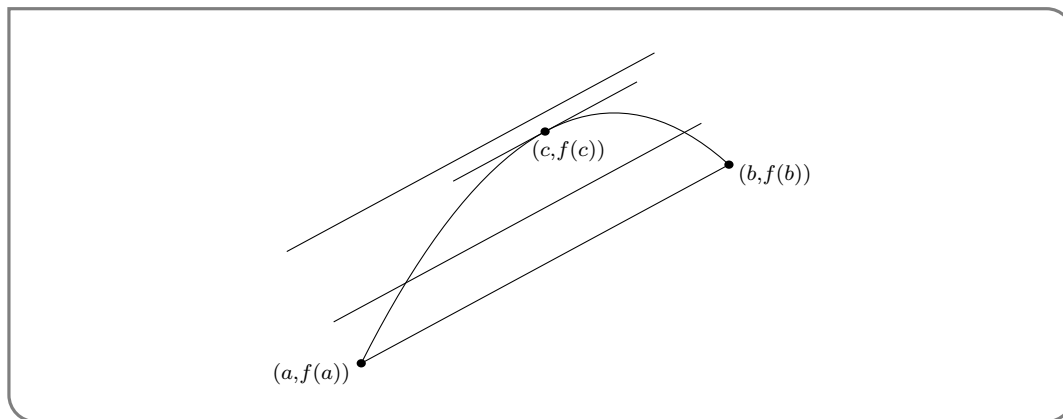
$$\begin{aligned} -1 &\leq f'(c) \leq 2 \\ -1 &\leq \frac{q - 10}{4} \leq 2 \\ -4 &\leq q - 10 \leq 8 \\ 6 &\leq q \leq 18 \end{aligned}$$

- Thus we must have  $6 \leq f(5) \leq 18$ .

Example 2.13.7

### ►►► (Optional) — Why is the MVT True?

We won't give a real proof for this theorem, but we'll look at a picture which shows why it is true. Here is the picture. It contains a sketch of the graph of  $f(x)$ , with  $x$  running from  $a$  to  $b$ , as well as a line segment which is the secant of the graph from the point  $(a, f(a))$  to the point  $(b, f(b))$ . The slope of the secant is exactly  $\frac{f(b)-f(a)}{b-a}$ . Remember that we are



looking for a point,  $(c, f(c))$ , on the graph of  $f(x)$  with the property that  $f'(c) = \frac{f(b)-f(a)}{b-a}$ , i.e. with the property that the slope of the tangent line at  $(c, f(c))$  is the same as the slope of the secant. So imagine that you start moving the secant upward, carefully keeping the moved line segment parallel to the secant. So the slope of the moved line segment is always exactly  $\frac{f(b)-f(a)}{b-a}$ . When we first start moving the line segment it is not tangent to the curve — it crosses the curve. This is illustrated in the figure by the second line segment from the bottom. If we move the line segment too far it does not touch the curve at all. This is illustrated in the figure by the top segment. But if we stop moving the line segment just before it stops intersecting the curve at all, we get exactly the tangent line to the curve at the point on the curve that is farthest from the secant. This tangent line has exactly the desired slope. This is illustrated in the figure by the third line segment from the bottom.

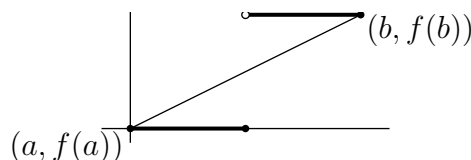
### ►► Be Careful with Hypotheses

The mean value theorem has hypotheses —  $f(x)$  has to be continuous for  $a \leq x \leq b$  and has to be differentiable for  $a < x < b$ . If either hypothesis is violated, the conclusion of the mean value theorem can fail. That is, the curve  $y = f(x)$  need not have a tangent line at some  $x = c$  between  $a$  and  $b$  whose slope,  $f'(c)$ , is the same as the slope,  $\frac{f(b)-f(a)}{b-a}$ , of the secant joining the points  $(a, f(a))$  and  $(b, f(b))$  on the curve. If  $f'(x)$  fails to exist for even a single value of  $x$  between  $a$  and  $b$ , all bets are off. The following two examples illustrate this.

#### Example 2.13.8

For the first “bad” example,  $a = 0$ ,  $b = 2$  and

$$f(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$



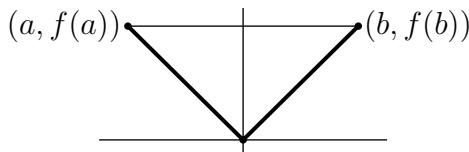
For this example,  $f'(x) = 0$  at every  $x$  where it is defined. That is, at every  $x \neq 1$ . But the slope of the secant joining  $(a, f(a)) = (0, 0)$  and  $(b, f(b)) = (2, 1)$  is  $\frac{1}{2}$ .

Example 2.13.8

Example 2.13.9

For the second “bad” example,  $a = -1$ ,  $b = 1$  and  $f(x) = |x|$ . For this function

$$f'(x) = \begin{cases} -1 & \text{if } x < 0 \\ \text{undefined} & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$



For this example,  $f'(x) = \pm 1$  at every  $x$  where it is defined. That is, at every  $x \neq 0$ . But the slope of the secant joining  $(a, f(a)) = (-1, 1)$  and  $(b, f(b)) = (1, 1)$  is 0.

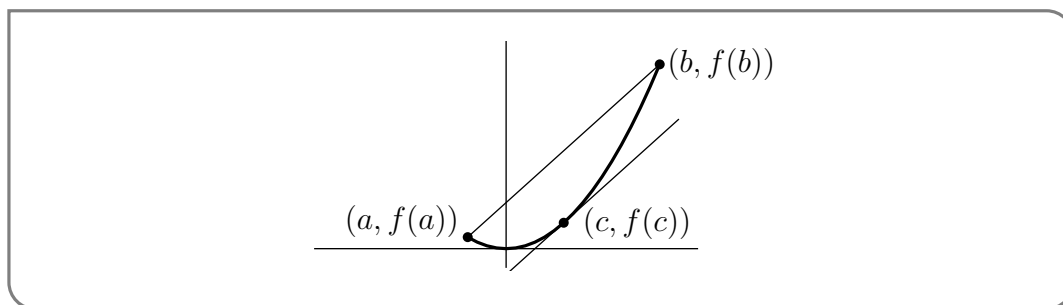
Example 2.13.9

Example 2.13.10

Here is one “good” example, where the hypotheses of the mean value theorem are satisfied. Let  $f(x) = x^2$ . Then  $f'(x) = 2x$ . For any  $a < b$ ,

$$\frac{f(b) - f(a)}{b - a} = \frac{b^2 - a^2}{b - a} = b + a$$

So  $f'(c) = 2c$  is exactly  $\frac{f(b) - f(a)}{b - a}$  when  $c = \frac{a+b}{2}$ , which, in this example, happens to be exactly half way between  $x = a$  and  $x = b$ .



Example 2.13.10

Recall from Section 2.3 that if  $f'(c) > 0$ , then  $f(x)$  is increasing at  $x = c$ . A simple consequence of the mean value theorem is that if you know the sign of  $f'(c)$  for all  $c$ 's between  $a$  and  $b$ , with  $b > a$ , then  $f(b) - f(a) = f'(c)(b - a)$  must have the same sign.

**Corollary 2.13.11** (Consequences of the mean value theorem).

Let  $A$  and  $B$  be real numbers with  $A < B$ . Let the function  $f(x)$  be defined and continuous on the closed interval  $A \leq x \leq B$  and be differentiable on the open interval  $A < x < B$ .

- (a) If  $f'(c) = 0$  for all  $A < c < B$ , then  $f(b) = f(a)$  for all  $A \leq a < b \leq B$ .  
— That is,  $f(x)$  is constant on  $A \leq x \leq B$ .
- (b) If  $f'(c) \geq 0$  for all  $A < c < B$ , then  $f(b) \geq f(a)$  for all  $A \leq a \leq b \leq B$ .  
— That is,  $f(x)$  is increasing on  $A \leq x \leq B$ .
- (c) If  $f'(c) > 0$  for all  $A < c < B$ , then  $f(b) > f(a)$  for all  $A \leq a < b \leq B$ .  
— That is,  $f(x)$  is strictly increasing on  $A \leq x \leq B$ .
- (d) If  $f'(c) \leq 0$  for all  $A < c < B$ , then  $f(b) \leq f(a)$  for all  $A \leq a \leq b \leq B$ .  
— That is,  $f(x)$  is decreasing on  $A \leq x \leq B$ .
- (e) If  $f'(c) < 0$  for all  $A < c < B$ , then  $f(b) < f(a)$  for all  $A \leq a < b \leq B$ .  
— That is,  $f(x)$  is strictly decreasing on  $A \leq x \leq B$ .

It is not hard to see why the above is true:

- Say  $f'(x) = 0$  at every point in the interval  $[A, B]$ . Now pick any  $a, b \in [A, B]$  with  $a < b$ . Then the MVT tells us that there is  $c \in (a, b)$  so that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

If  $f(b) \neq f(a)$  then we must have that  $f'(c) \neq 0$  — contradicting what we are told about  $f'(x)$ . Thus we must have that  $f(b) = f(a)$ .

- Similarly, say  $f'(x) \geq 0$  at every point in the interval  $[A, B]$ . Now pick any  $a, b \in [A, B]$  with  $a < b$ . Then the MVT tells us that there is  $c \in (a, b)$  so that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Since  $b > a$ , the denominator is positive. Now if  $f(b) < f(a)$  the numerator would be negative, making the right-hand side negative, and contradicting what we are told about  $f'(x)$ . Hence we must have  $f(b) \geq f(a)$ .

- The other parts are similar.

A nice corollary of the above corollary is the following:

**Corollary 2.13.12.**

If  $f'(x) = g'(x)$  for all  $x$  in the open interval  $(a, b)$ , then  $f - g$  is a constant on  $(a, b)$ . That is  $f(x) = g(x) + c$ , where  $c$  is some constant.

We can prove this by setting  $h(x) = f(x) - g(x)$ . Then  $h'(x) = 0$  and so the previous corollary tells us that  $h(x)$  is constant.

**Example 2.13.13**

Using this corollary we can prove results like the following:

$$\arcsin x + \arccos x = \frac{\pi}{2} \quad \text{for all } -1 < x < 1$$

How does this work? Let  $f(x) = \arcsin x + \arccos x$ . Then

$$f'(x) = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$$

Thus  $f$  must be a constant. To find out which constant, we can just check its value at a convenient point, like  $x = 0$ .

$$\arcsin(0) + \arccos(0) = \pi/2 + 0 = \pi/2$$

Since the function is constant, this must be the value.

**Example 2.13.13**

## 2.14 ▲ Higher Order Derivatives

The operation of differentiation takes as input one function,  $f(x)$ , and produces as output another function,  $f'(x)$ . Now  $f'(x)$  is once again a function. So we can differentiate it again, assuming that it is differentiable, to create a third function, called the second derivative of  $f$ . And we can differentiate the second derivative again to create a fourth function, called the third derivative of  $f$ . And so on.

**Notation 2.14.1.**

- $f''(x)$  and  $f^{(2)}(x)$  and  $\frac{d^2 f}{dx^2}(x)$  all mean  $\frac{d}{dx}\left(\frac{d}{dx}f(x)\right)$
- $f'''(x)$  and  $f^{(3)}(x)$  and  $\frac{d^3 f}{dx^3}(x)$  all mean  $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}f(x)\right)\right)$
- $f^{(4)}(x)$  and  $\frac{d^4 f}{dx^4}(x)$  both mean  $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}f(x)\right)\right)\right)$
- and so on.

Here is a simple example. Then we'll think a little about the significance of second order derivatives. Then we'll do a more a computationally complex example.

## Example 2.14.2

Let  $n$  be a natural number and let  $f(x) = x^n$ . Then

$$\begin{aligned}\frac{d}{dx}x^n &= nx^{n-1} \\ \frac{d^2}{dx^2}x^n &= \frac{d}{dx}(nx^{n-1}) = n(n-1)x^{n-2} \\ \frac{d^3}{dx^3}x^n &= \frac{d}{dx}(n(n-1)x^{n-2}) = n(n-1)(n-2)x^{n-3}\end{aligned}$$

Each time we differentiate, we bring down the exponent, which is exactly one smaller than the previous exponent brought down, and we reduce the exponent by one. By the time we have differentiated  $n-1$  times, the exponent has decreased to  $n - (n-1) = 1$  and we have brought down the factors  $n(n-1)(n-2) \cdots 2$ . So

$$\frac{d^{n-1}}{dx^{n-1}}x^n = n(n-1)(n-2) \cdots 2x$$

and

$$\frac{d^n}{dx^n}x^n = n(n-1)(n-2) \cdots 1$$

The product of the first  $n$  natural numbers,  $1 \cdot 2 \cdot 3 \cdots n$ , is called “ $n$  factorial” and is denoted  $n!$ . So we can also write

$$\frac{d^n}{dx^n}x^n = n!$$

If  $m > n$ , then

$$\frac{d^m}{dx^m}x^n = 0$$

## Example 2.14.2

## Example 2.14.3

Recall that the derivative  $v'(a)$  is the (instantaneous) rate of change of the function  $v(t)$  at  $t = a$ . Suppose that you are walking on the  $x$ -axis and that  $x(t)$  is your  $x$ -coordinate at time  $t$ . Also suppose, for simplicity, that you are moving from left to right. Then  $v(t) = x'(t)$  is your velocity at time  $t$  and  $v'(a) = x''(a)$  is the rate at which your velocity is changing at time  $t = a$ . It is called your acceleration. In particular, if  $x''(a) > 0$ , then your velocity is increasing, i.e. you are speeding up, at time  $a$ . If  $x''(a) < 0$ , then your velocity is decreasing, i.e. you are slowing down, at time  $a$ . That's one interpretation of the second derivative.

## Example 2.14.3

Example 2.14.4 (Example 2.11.1, continued)

Find  $y''$  if  $y = y^3 + xy + x^3$ .

*Solution.* This problem concerns some function  $y(x)$  that is not given to us explicitly. All that we are told is that  $y(x)$  satisfies

$$y(x) = y(x)^3 + xy(x) + x^3 \quad (\text{E1})$$

for all  $x$ . We are asked to find  $y''(x)$ . We cannot solve this equation to get an explicit formula for  $y(x)$ . So we use implicit differentiation, as we did in Example 2.11.1. That is, we apply  $\frac{d}{dx}$  to both sides of (E1). This gives

$$y'(x) = 3y(x)^2 y'(x) + y(x) + x y'(x) + 3x^2 \quad (\text{E2})$$

which we can solve for  $y'(x)$ , by moving all  $y'(x)$ 's to the left hand side, giving

$$[1 - x - 3y(x)^2] y'(x) = y(x) + 3x^2$$

and then dividing across.

$$y'(x) = \frac{y(x) + 3x^2}{1 - x - 3y(x)^2} \quad (\text{E3})$$

To get  $y''(x)$ , we have two options.

*Method 1.* Apply  $\frac{d}{dx}$  to both sides of (E2). This gives

$$y''(x) = 3y(x)^2 y''(x) + 6y(x) y'(x)^2 + 2y'(x) + x y''(x) + 6x$$

We can now solve for  $y''(x)$ , giving

$$y''(x) = \frac{6x + 2y'(x) + 6y(x)y'(x)^2}{1 - x - 3y(x)^2} \quad (\text{E4})$$

Then we can substitute in (E3), giving

$$\begin{aligned} y''(x) &= 2 \frac{3x + \frac{y(x)+3x^2}{1-x-3y(x)^2} + 3y(x) \left( \frac{y(x)+3x^2}{1-x-3y(x)^2} \right)^2}{1 - x - 3y(x)^2} \\ &= 2 \frac{3x[1 - x - 3y(x)^2]^2 + [y(x) + 3x^2][1 - x - 3y(x)^2] + 3y(x)[y(x) + 3x^2]^2}{[1 - x - 3y(x)^2]^3} \end{aligned}$$

*Method 2.* Alternatively, we can also differentiate (E3).

$$\begin{aligned} y''(x) &= \frac{[y'(x) + 6x][1 - x - 3y(x)^2] - [y(x) + 3x^2][-1 - 6y(x)y'(x)]}{[1 - x - 3y(x)^2]^2} \\ &= \frac{\left[ \frac{y(x)+3x^2}{1-x-3y(x)^2} + 6x \right][1 - x - 3y(x)^2] - [y(x) + 3x^2][-1 - 6y(x) \frac{y(x)+3x^2}{1-x-3y(x)^2}]}{[1 - x - 3y(x)^2]^2} \\ &= \frac{2[y(x) + 3x^2][1 - x - 3y(x)^2] + 6x[1 - x - 3y(x)^2]^2 + 6y(x)[y(x) + 3x^2]^2}{[1 - x - 3y(x)^2]^3} \end{aligned}$$

*Remark 1.* We have now computed  $y''(x)$  — sort of. The answer is in terms of  $y(x)$ , which we don't know. Since we cannot get an explicit formula for  $y(x)$ , there's not a great deal that we can do, in general.

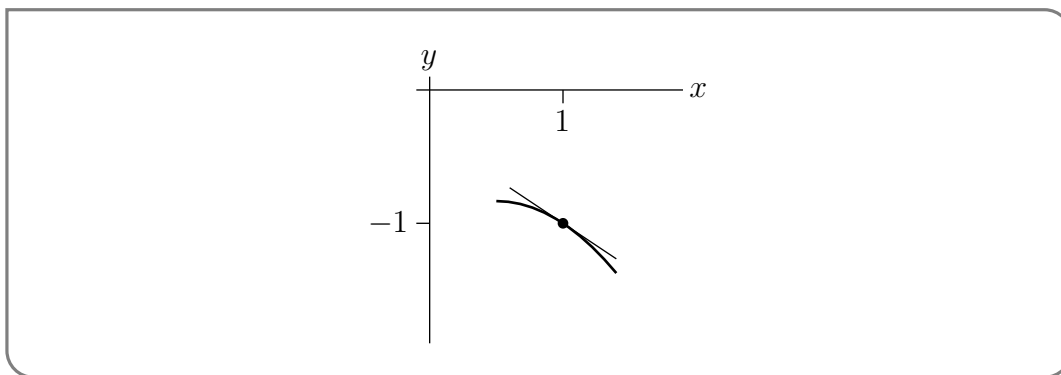
*Remark 2.* Even though we cannot solve  $y = y^3 + xy + x^3$  explicitly for  $y(x)$ , for general  $x$ , it is sometimes possible to solve equations like this for some special values of  $x$ . In fact, we saw in Example 2.11.1 that when  $x = 1$ , the given equation reduces to  $y(1) = y(1)^3 + 1 \cdot y(1) + 1^3$ , or  $y(1)^3 = -1$ , which we can solve to get  $y(1) = -1$ . Substituting into (E2), as we did in Example 2.11.1 gives

$$y'(1) = \frac{-1 + 3}{1 - 1 - 3(-1)^2} = -\frac{2}{3}$$

and substituting into (E4) gives

$$y''(1) = \frac{6 + 2(-\frac{2}{3}) + 6(-1)(-\frac{2}{3})^2}{1 - 1 - 3(-1)^2} = \frac{6 - \frac{4}{3} - \frac{8}{3}}{-3} = -\frac{2}{3}$$

(It's a fluke that, in this example,  $y'(1)$  and  $y''(1)$  happen to be equal.) So we now know that, even though we can't solve  $y = y^3 + xy + x^3$  explicitly for  $y(x)$ , the graph of the solution passes through  $(1, -1)$  and has slope  $-\frac{2}{3}$  (i.e. is sloping downwards by between  $30^\circ$  and  $45^\circ$ ) there and, furthermore, the slope of the graph decreases as  $x$  increases through  $x = 1$ .



Here is a sketch of the part of the graph very near  $(1, -1)$ . The tangent line to the graph at  $(1, -1)$  is also shown. Note that the tangent line is sloping down to the right, as we expect, and that the graph lies below the tangent line near  $(1, -1)$ . That's because the slope  $f'(x)$  is decreasing (becoming more negative) as  $x$  passes through 1.

Example 2.14.4

#### Warning 2.14.5.

Many people will suppress the  $(x)$  in  $y(x)$  when doing computations like those in Example 2.14.4. This gives shorter, easier to read formulae, like  $y' = \frac{y+3x^2}{1-x-3y^2}$ . If you do this, you must never forget that  $y$  is a function of  $x$  and is *not* a constant. If you do forget, you'll make the very serious error of saying that  $\frac{dy}{dx} = 0$ , which is false.

## 2.15 ▲ (Optional) — Is $\lim_{x \rightarrow c} f'(x)$ Equal to $f'(c)$ ?

Consider the function

$$f(x) = \begin{cases} \frac{\sin x^2}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

For any  $x \neq 0$  we can easily use our differentiation rules to find

$$f'(x) = \frac{2x^2 \cos x^2 - \sin x^2}{x^2}$$

But for  $x = 0$  none of our usual differentiation rules apply. So how do we find  $f'(0)$ ? One obviously legitimate strategy is to directly apply the Definition 2.2.1 of the derivative. As an alternative, we will now consider the question “Can one find  $f'(0)$  by taking the limit of  $f'(x)$  as  $x$  tends to zero?”. There is bad news and there is good news.

- The bad news is that, even for functions  $f(x)$  that are differentiable for all  $x$ ,  $f'(x)$  need not be continuous. That is, it is *not* always true that  $\lim_{x \rightarrow 0} f'(x) = f'(0)$ . We will see a function for which  $\lim_{x \rightarrow 0} f'(x) \neq f'(0)$  in Example 2.15.1 below.
- The good news is that Theorem 2.15.2 below provides conditions which are sufficient to guarantee that  $f(x)$  is differentiable at  $x = 0$  and that  $\lim_{x \rightarrow 0} f'(x) = f'(0)$ .

### Example 2.15.1

Consider the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

For  $x \neq 0$  we have, by the product and chain rules,

$$\begin{aligned} f'(x) &= 2x \sin \frac{1}{x} + x^2 \left( \cos \frac{1}{x} \right) \left( -\frac{1}{x^2} \right) \\ &= 2x \sin \frac{1}{x} - \cos \frac{1}{x} \end{aligned}$$

As  $\left| \sin \frac{1}{x} \right| \leq 1$ , we have

$$\lim_{x \rightarrow 0} 2x \sin \frac{1}{x} = 0$$

On the other hand, as  $x$  tends to zero,  $\frac{1}{x}$  goes to  $\pm\infty$ . So

$$\lim_{x \rightarrow 0} \cos \frac{1}{x} = DNE \implies \lim_{x \rightarrow 0} f'(x) = DNE$$

We will now see that, despite this,  $f'(0)$  is perfectly well defined. By definition

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h} \\ &= \lim_{h \rightarrow 0} h \sin \frac{1}{h} \\ &= 0 \quad \text{since } \left| \sin \frac{1}{h} \right| \leq 1 \end{aligned}$$

So  $f'(0)$  exists, but is not equal to  $\lim_{x \rightarrow 0} f'(x)$ , which does not exist.

Example 2.15.1

Now for the good news.

**Theorem 2.15.2.**

Let  $a < c < b$ . Assume that

- the function  $f(x)$  is continuous on the interval  $a < x < b$  and
- is differentiable at every  $x$  in the intervals  $a < x < c$  and  $c < x < b$  and
- the limit  $\lim_{x \rightarrow c} f'(x)$  exists.

Then  $f$  is differentiable at  $x = c$  and

$$f'(c) = \lim_{x \rightarrow c} f'(x)$$

*Proof.* By hypothesis, there is a number  $L$  such that

$$\lim_{x \rightarrow c} f'(x) = L$$

By definition

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

By the Mean Value Theorem (Theorem 2.13.4) there is, for each  $h$ , an (unknown) number  $x_h$  between  $c$  and  $c+h$  such that  $f'(x_h) = \frac{f(c+h) - f(c)}{h}$ . So

$$f'(c) = \lim_{h \rightarrow 0} f'(x_h)$$

As  $h$  tends to zero,  $c+h$  tends to  $c$ , and so  $x_h$  is forced to tend to  $c$ , and  $f'(x_h)$  is forced to tend to  $L$  so that

$$f'(c) = \lim_{h \rightarrow 0} f'(x_h) = L$$

□

In the next example we evaluate  $f'(0)$  by applying Theorem 2.15.2

Example 2.15.3

Let

$$f(x) = \begin{cases} \frac{\sin x^2}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

We have already observed above that, for  $x \neq 0$ ,

$$f'(x) = \frac{2x^2 \cos x^2 - \sin x^2}{x^2} = 2 \cos x^2 - \frac{\sin x^2}{x^2}$$

We use Theorem 2.15.2 with  $c = 0$  to show that  $f(x)$  is differentiable at  $x = 0$  and to evaluate  $f'(0)$ . That theorem has two hypotheses that we have not yet verified, namely the continuity of  $f(x)$  at  $x = 0$ , and the existence of the limit  $\lim_{x \rightarrow 0} f'(x)$ . We verify them now.

- We already know, by Lemma 2.8.1 that  $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$ . So

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} &= \lim_{h \rightarrow 0^+} \frac{\sin h}{h} \quad \text{with } h = x^2 \\ &= 1 \end{aligned}$$

and

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x^2}{x} = \lim_{x \rightarrow 0} x \frac{\sin x^2}{x^2} = \lim_{x \rightarrow 0} x \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = 0 \times 1 = 0$$

and  $f(x)$  is continuous at  $x = 0$ .

- The limit of the derivative is

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \left[ 2 \cos x^2 - \frac{\sin x^2}{x^2} \right] = 2 \times 1 - 1 = 1$$

So, by Theorem 2.15.2,  $f(x)$  is differentiable at  $x = 0$  and  $f'(0) = 1$ .

Example 2.15.3

Q[27]: Suppose  $f^{-1}(4x - 1) = \frac{2x + 3}{x + 1}$ . Evaluate  $f(0)$ .

Q[28]: Suppose a curve is defined implicitly by

$$\arcsin(x + 2y) = x^2 + y^2$$

Solve for  $y'$  in terms of  $x$  and  $y$ .

## 2.13▲ The Mean Value Theorem

### Exercises

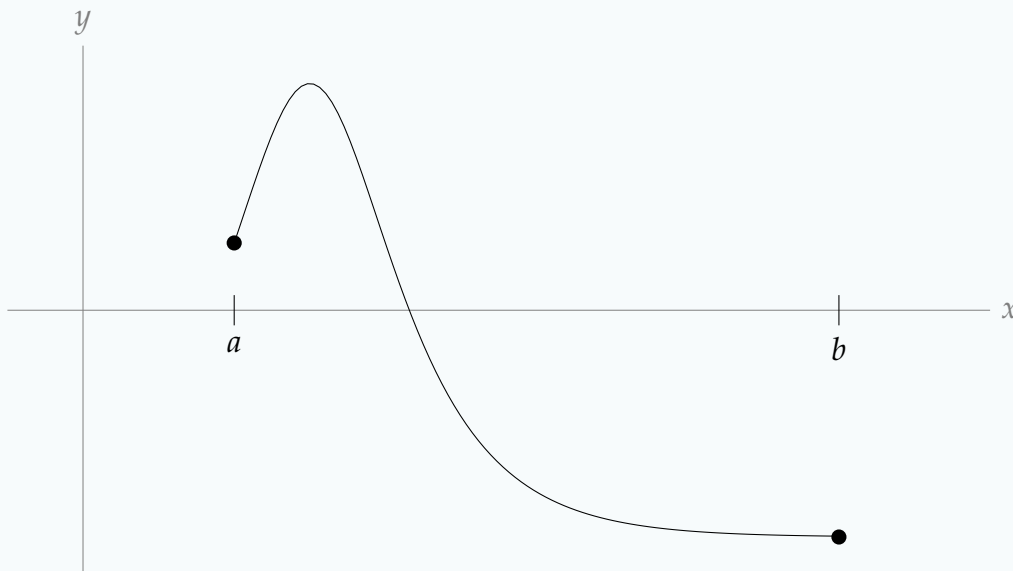
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#### ► Stage 1

Q[1]: Suppose a particular caribou has a top speed of 70 kph, and in one year it migrates 5000 km. What do you know about the amount of time the caribou spent travelling during its migration?

Q[2]: Suppose a migrating sandhill crane flew 240 kilometres in one day. What does the mean value theorem tell you about its speed during that day?

Q[3]: Below is the graph of  $y = f(x)$ , where  $x$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Mark on the graph the approximate location of a value  $c$  guaranteed by the mean value theorem.



Q[4]: Give a function  $f(x)$  with the properties that:

- $f(x)$  is differentiable on the open interval  $0 < x < 10$

- $f(0) = 0, f(10) = 10$

but for all  $c \in (0, 10)$ ,  $f'(c) = 0$ .

Q[5]: For each of the parts below, sketch a function  $f(x)$  (different in each part) that is continuous and differentiable over all real numbers, with  $f(1) = f(2) = 0$ , and with the listed property, or explain why no such function exists.

- (a)  $f'(c) = 0$  for *no* point  $c \in (1, 2)$
- (b)  $f'(c) = 0$  for *exactly one* point  $c \in (1, 2)$
- (c)  $f'(c) = 0$  for *exactly five* points  $c \in (1, 2)$
- (d)  $f'(c) = 0$  for *infinitely many* points  $c \in (1, 2)$

Q[6]: Suppose you want to show that a point exists where the function  $f(x) = \sqrt{|x|}$  has a tangent line with slope  $\frac{1}{13}$ . Find the mistake(s) in the following work, and give a correct proof.

The function  $f(x)$  is continuous and differentiable over all real numbers, so the mean value theorem applies.  $f(-4) = 2$  and  $f(9) = 3$ , so by the mean value theorem, there exists some  $c \in (-4, 9)$  such that  $f'(x) = \frac{3-2}{9-(-4)} = \frac{1}{13}$ .

## ► Stage 2

Q[7](\*) : Let  $f(x) = x^2 - 2\pi x + \cos(x) - 1$ . Show that there exists a real number  $c$  such that  $f'(c) = 0$ .

Q[8](\*) : Let  $f(x) = e^x + (1 - e)x^2 - 1$ . Show that there exists a real number  $c$  such that  $f'(c) = 0$ .

Q[9](\*) : Let  $f(x) = \sqrt{3 + \sin(x)} + (x - \pi)^2$ . Show that there exists a real number  $c$  such that  $f'(c) = 0$ .

Q[10](\*) : Let  $f(x) = x \cos(x) - x \sin(x)$ . Show that there exists a real number  $c$  such that  $f'(c) = 0$ .

Q[11]: How many roots does the function  $f(x) = 3x - \sin x$  have?

Q[12]: How many roots does the function  $f(x) = \frac{(4x+1)^4}{16} + x$  have?

Q[13]: How many roots does the function  $f(x) = x^3 + \sin(x^5)$  have?

Q[14]: How many positive-valued solutions does the equation  $e^x = 4 \cos(2x)$  have?

Q[15](\*) : Let  $f(x) = 3x^5 - 10x^3 + 15x + a$ , where  $a$  is some constant.

- (a) Prove that, regardless of the value  $a$ ,  $f'(x) > 0$  for all  $x$  in  $(-1, 1)$ .  
 (b) Prove that, regardless of the value  $a$ ,  $f(x) = 3x^5 - 10x^3 + 15x + a$  has at most one root in  $[-1, 1]$ .

Q[16](\*) : Find the point promised by the Mean Value Theorem for the function  $e^x$  on the interval  $[0, T]$ .

Q[17]: Use Corollary 2.13.11 and Theorem 2.12.7 in the CLP-1 text to show that  $\operatorname{arcsec} x = C - \operatorname{arccsc} x$  for some constant  $C$ . Find  $C$ .

### ► Stage 3

Q[18](\*) : Suppose  $f(0) = 0$  and  $f'(x) = \frac{1}{1 + e^{-f(x)}}$ . Prove that  $f(100) < 100$ .

Remark: an equation relating a function to its own derivative is called a differential equation. We'll see some very basic differential equations in Section 3.3.

Q[19]: Let  $f(x) = 2x + \sin x$ . What is the largest interval containing  $x = 0$  over which  $f(x)$  is one-to-one? What are the domain and range of  $f^{-1}(x)$ ?

Q[20]: Let  $f(x) = \frac{x}{2} + \sin x$ . What is the largest interval containing  $x = 0$  over which  $f(x)$  is one-to-one? What are the domain and range of  $f^{-1}(x)$ , if we restrict  $f$  to this interval?

Q[21]: Suppose  $f(x)$  and  $g(x)$  are functions that are continuous over the interval  $[a, b]$  and differentiable over the interval  $(a, b)$ . Suppose further that  $f(a) < g(a)$  and  $g(b) < f(b)$ . Show that there exists some  $c \in [a, b]$  with  $f'(c) > g'(c)$ .

Q[22]: Suppose  $f(x)$  is a function that is differentiable over all real numbers, and  $f'(x)$  has precisely two roots. What is the maximum number of distinct roots that  $f(x)$  may have?

Q[23]: How many roots does  $f(x) = \sin x + x^2 + 5x + 1$  have?

## 2.14▲ Higher Order Derivatives

### Exercises

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### ► Stage 1

Q[1]: What is the 180th derivative of the function  $f(x) = e^x$ ?

Q[2]: Suppose  $f(x)$  is a differentiable function, with  $f'(x) > 0$  and  $f''(x) > 0$  for every  $x \in (a, b)$ . Which of the following must be true?

- (i)  $f(x)$  is positive over  $(a, b)$
- (ii)  $f(x)$  is increasing over  $(a, b)$
- (iii)  $f(x)$  is increasing at a constant rate over  $(a, b)$
- (iv)  $f(x)$  is increasing faster and faster over  $(a, b)$
- (v)  $f'''(x) > 0$  for some  $x \in (a, b)$

Q[3]: Let  $f(x) = ax^{15}$  for some constant  $a$ . Which value of  $a$  results in  $f^{(15)}(x) = 3$ ?

Q[4]: Find the mistake(s) in the following work, and provide a corrected answer.

Suppose  $-14x^2 + 2xy + y^2 = 1$ . We find  $\frac{d^2y}{dx^2}$  at the point  $(1, 3)$ . Differentiating implicitly:

$$-28x + 2y + 2xy' + 2yy' = 0$$

Plugging in  $x = 1, y = 3$ :

$$\begin{aligned} -28 + 6 + 2y' + 6y' &= 0 \\ y' &= \frac{11}{4} \end{aligned}$$

Differentiating:

$$y'' = 0$$

## ► Stage 2

Q[5]: Let  $f(x) = (\log x - 1)x$ . Evaluate  $f''(x)$ .

Q[6]: Evaluate  $\frac{d^2}{dx^2} \{\arctan x\}$ .

Q[7]: The unit circle consists of all point  $x^2 + y^2 = 1$ . Give an expression for  $\frac{d^2y}{dx^2}$  in terms of  $y$ .

Q[8]: Suppose the position of a particle at time  $t$  is given by  $s(t) = \frac{e^t}{t^2 + 1}$ . Find the acceleration of the particle at time  $t = 1$ .

Q[9]: Evaluate  $\frac{d^3}{dx^3} \{\log(5x^2 - 12)\}$ .

Q[10]: The height of a particle at time  $t$  seconds is given by  $h(t) = -\cos t$ . Is the particle

speeding up or slowing down at  $t = 1$ ?

Q[11]: The height of a particle at time  $t$  seconds is given by  $h(t) = t^3 - t^2 - 5t + 10$ . Is the particle's motion getting faster or slower at  $t = 1$ ?

Q[12]: Suppose a curve is defined implicitly by

$$x^2 + x + y = \sin(xy)$$

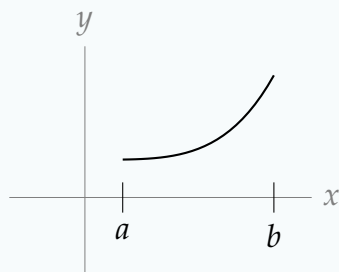
What is  $\frac{d^2y}{dx^2}$  at the point  $(0, 0)$ ?

Q[13]: Which statements below are true, and which false?

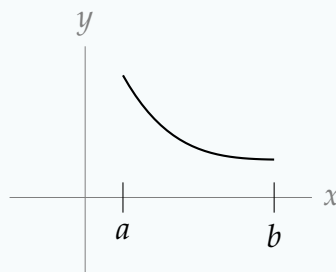
- (a)  $\frac{d^4}{dx^4} \sin x = \sin x$
- (b)  $\frac{d^4}{dx^4} \cos x = \cos x$
- (c)  $\frac{d^4}{dx^4} \tan x = \tan x$

### ► Stage 3

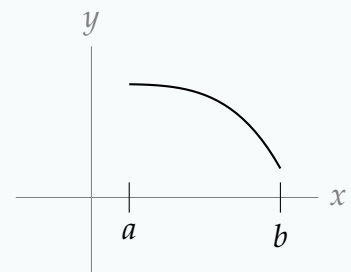
Q[14]: A function  $f(x)$  satisfies  $f'(x) < 0$  and  $f''(x) > 0$  over  $(a, b)$ . Which of the following curves below might represent  $y = f(x)$ ?



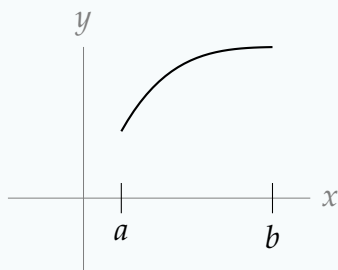
(i)



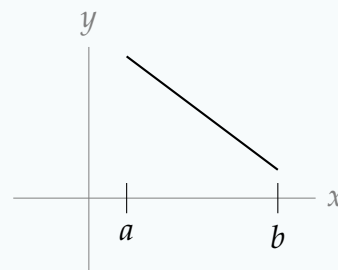
(ii)



(iii)



(iv)



(v)

Q[15]: Let  $f(x) = 2^x$ . What is  $f^{(n)}(x)$ , if  $n$  is a whole number?

Q[16]: Let  $f(x) = ax^3 + bx^2 + cx + d$ , where  $a, b, c$ , and  $d$  are nonzero constants. What is the smallest integer  $n$  so that  $\frac{d^n f}{dx^n} = 0$  for all  $x$ ?

Q[17](\*):

$$f(x) = e^{x+x^2}$$

$$h(x) = 1 + x + \frac{3}{2}x^2$$

- (a) Find the first and second derivatives of both functions
- (b) Evaluate both functions and their first and second derivatives at 0.
- (c) Show that for all  $x > 0$ ,  $f(x) > h(x)$ .

Remark: for some applications, we only need to know that a function is “big enough.” Since  $f(x)$  is a difficult function to evaluate, it may be useful to know that it is bigger than  $h(x)$  when  $x$  is positive.

Q[18](\*): The equation  $x^3y + y^3 = 10x$  defines  $y$  implicitly as a function of  $x$  near the point  $(1, 2)$ .

- (a) Compute  $y'$  at this point.
- (b) It can be shown that  $y''$  is negative when  $x = 1$ . Use this fact and your answer to (a) to make a sketch showing the relationship of the curve to its tangent line at  $(1, 2)$ .

Q[19]: Let  $g(x) = f(x)e^x$ . In Question 12, Section 2.7, we learned that  $g'(x) = [f(x) + f'(x)]e^x$ .

- (a) What is  $g''(x)$ ?
- (b) What is  $g'''(x)$ ?
- (c) Based on your answers above, guess a formula for  $g^{(4)}(x)$ . Check it by differentiating.

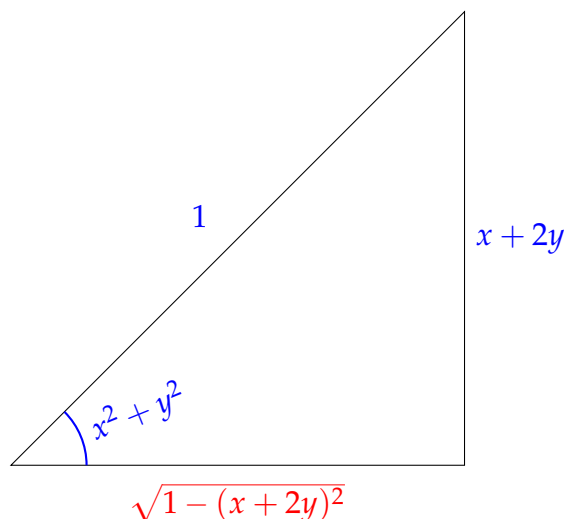
Q[20]: Suppose  $f(x)$  is a function whose first  $n$  derivatives exist over all real numbers, and  $f^{(n)}(x)$  has precisely  $m$  roots. What is the maximum number of roots that  $f(x)$  may have?

Q[21]: How many roots does the function  $f(x) = (x + 1)\log(x + 1) + \sin x - x^2$  have?

Q[22](\*): Let  $f(x) = x|x|$ .

- (a) Show that  $f(x)$  is differentiable at  $x = 0$ , and find  $f'(0)$ .
- (b) Find the second derivative of  $f(x)$ . Explicitly state, with justification, the point(s) at which  $f''(x)$  does not exist, if any.





where the adjacent side (in red) come from the Pythagorean Theorem. Then,  $\cos(x^2 + y^2) = \sqrt{1 - (x + 2y)^2}$ , so using our second solution:

$$\begin{aligned} y' &= \frac{2x \cos(x^2 + y^2) - 1}{2 - 2y \cos(x^2 + y^2)} \\ &= \frac{2x \sqrt{1 - (x + 2y)^2} - 1}{2 - 2y \sqrt{1 - (x + 2y)^2}} \end{aligned}$$

which is exactly the answer from our first solution.

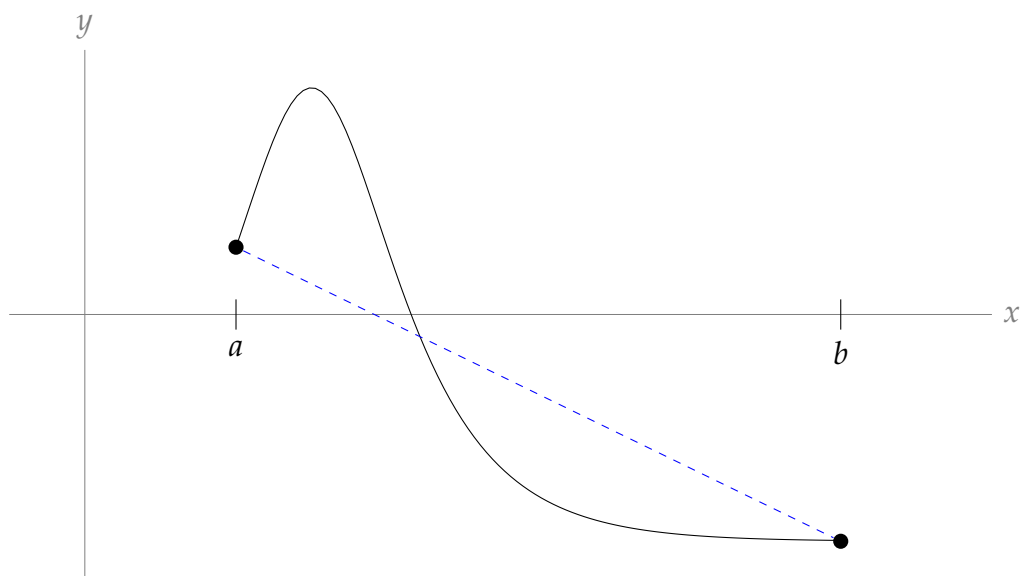
## Solutions to Exercises [2.13](#) — Jump to [TABLE OF CONTENTS](#)

**S-1:** We know the top speed of the caribou, so we can use this to give the *minimum* possible number of hours the caribou spent travelling during its migration. If the caribou travels at 70kph, it will take  $(5000\text{km}) \left( \frac{1\text{hr}}{70\text{km}} \right) \approx 71.4\text{hrs}$  to travel 5000 kilometres.

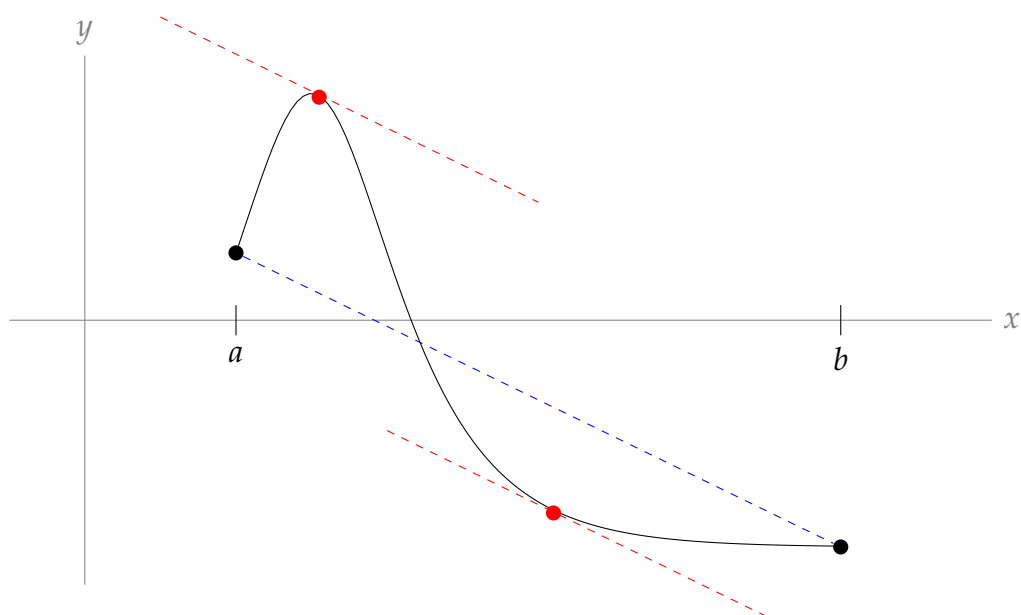
Probably the caribou wasn't sprinting the whole time, so probably it took it longer than that, but we can only say for sure that the caribou spent *at least* about 71.4 hours migrating.

**S-2:** If  $f(x)$  is the position of the crane at time  $x$ , measured in hours, then (if we let  $x = 0$  be the beginning of the day) we know that  $f(24) - f(0) = 240$ . Since  $f(x)$  is the position of the bird,  $f(x)$  is continuous and differentiable. So, the MVT says there is a  $c$  in  $(0, 24)$  such that  $f'(c) = \frac{f(24) - f(0)}{24 - 0} = \frac{240}{24} = 10$ . That is, at some point  $c$  during the day, the speed of the crane was exactly 10 kph.

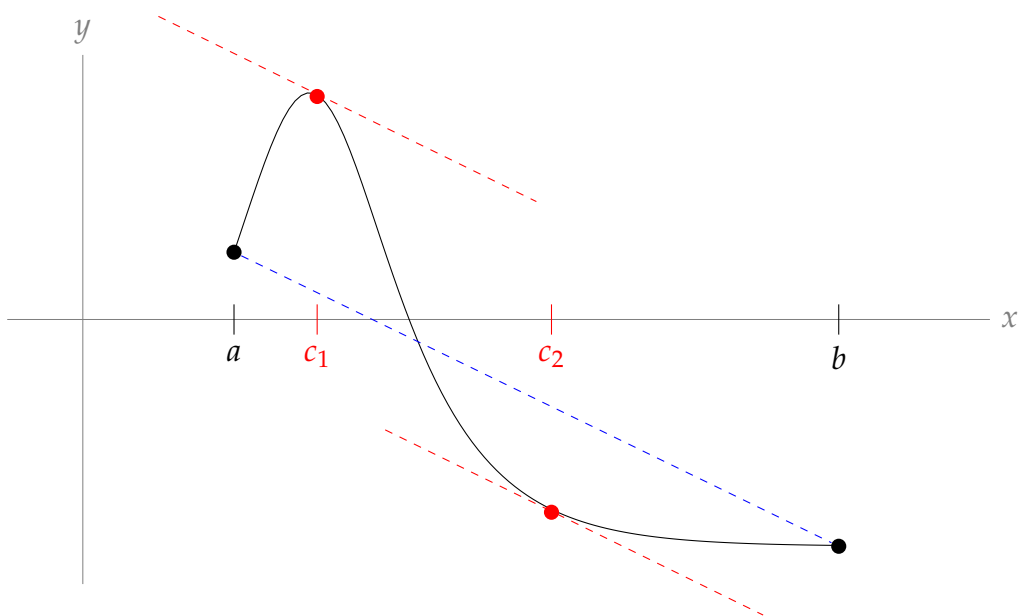
**S-3:** The MVT guarantees there is some point  $c$  strictly between  $a$  and  $b$  where the tangent line to  $f(x)$  at  $x = c$  has the same slope as the secant line of  $f(x)$  from  $x = a$  to  $x = b$ . So, let's start by drawing in the secant line.



What we're looking for is a point on the curve where the tangent line is parallel to this secant line. In fact, there are two.

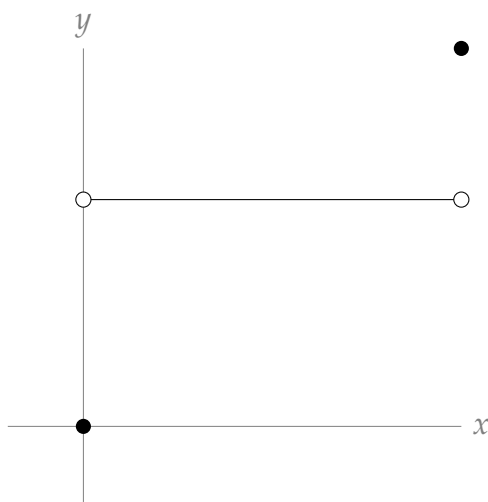


So, either of the two values  $c_1$  and  $c_2$  marked below can serve as the point guaranteed by the MVT:



**S-4:** Since  $f(x)$  is differentiable for all  $x \in (0, 10)$ , then  $f(x)$  is also continuous for all  $x \in (0, 10)$ . If  $f(x)$  were continuous on the closed interval  $[0, 10]$ , then the MVT would guarantee  $f'(x) = \frac{f(10) - f(0)}{10 - 0} = 1$  for some  $c \in (0, 10)$ ; however, this is not the case. So, it must be that  $f(x)$  is continuous for all  $x \in (0, 10)$ , but not for all  $x \in [0, 10]$ .

Since  $f'(c) = 0$  for  $c \in (0, 10)$ , that means  $f$  is constant on that interval. So,  $f(x)$  is a function like this:



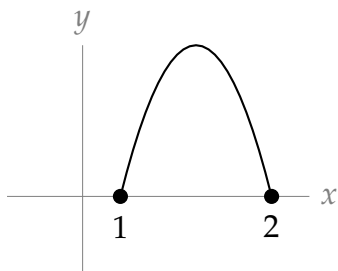
where the height of the constant function can be anything.

So, one possible answer is  $f(x) = \begin{cases} 0 & x \neq 10 \\ 10 & x = 10 \end{cases}$ .

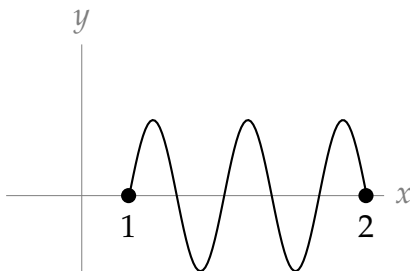
**S-5:** (a) No such function is possible: Rolle's Theorem guarantees  $f'(c) = 0$  for at least one point  $c \in (1, 2)$ .

For the other functions, examples are below, but many answers are possible.

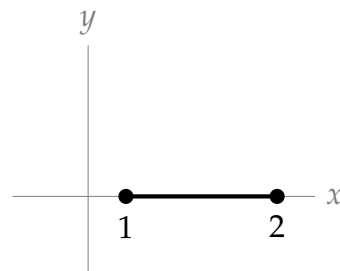
(b)



(c)



(d)



**S-6:** The function  $f(x)$  is continuous over all real numbers, but it is only differentiable when  $x \neq 0$ . So, if we want to apply the MVT, our interval must consist of only positive numbers or only negative numbers: the interval  $(-4, 13)$  is not valid.

It is possible to use the mean value theorem to prove what we want: if  $a = 1$  and  $b = 144$ , then  $f(x)$  is differentiable over the interval  $(1, 144)$  (since 0 is not contained in that interval), and  $f(x)$  is continuous everywhere, so by the mean value theorem there exists

some point  $c$  where  $f'(x) = \frac{\sqrt{|144|} - \sqrt{|1|}}{144 - 1} = \frac{11}{143} = \frac{1}{13}$ .

That being said, an easier way to prove that a point exists is to simply find it—without using the MVT. When  $x > 0$ ,  $f(x) = \sqrt{x}$ , so  $f'(x) = \frac{1}{2\sqrt{x}}$ . Then  $f'\left(\frac{169}{4}\right) = \frac{1}{13}$ .

**S-7:** We note that  $f(0) = f(2\pi) = 0$ . Then using the Mean Value Theorem (note that the function is differentiable for all real numbers), we conclude that there exists  $c$  in  $(0, 2\pi)$  such that

$$f'(c) = \frac{f(2\pi) - f(0)}{2\pi - 0} = 0.$$

**S-8:** We note that  $f(0) = f(1) = 0$ . Then using the Mean Value Theorem (note that the function is differentiable for all real numbers), we get that there exists  $c \in (0, 1)$  such that

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = 0.$$

**S-9:** We note that  $f(0) = f(2\pi) = \sqrt{3} + \pi^2$ . Then using the Mean Value Theorem (note that the function is differentiable for all real numbers since  $3 + \sin x > 0$ ), we get that there exists  $c \in (0, 2\pi)$  such that

$$f'(c) = \frac{f(2\pi) - f(0)}{2\pi - 0} = 0.$$

---

**S-10:** We note that  $f(0) = 0$  and  $f(\pi/4) = 0$ . Then using the Mean Value Theorem (note that the function is differentiable for all real numbers), we get that there exists  $c \in (0, \pi/4)$  such that

$$f'(c) = \frac{f(\pi/4) - f(0)}{\pi/4 - 0} = 0.$$

**S-11:** By inspection, we see  $x = 0$  is a root of  $f(x)$ . The question now is whether there could possibly be other roots. Since  $f(x)$  is differentiable over all real numbers, if there is another root  $a$ , then by Rolle's Theorem,  $f'(c) = 0$  for some  $c$  strictly between 0 and  $a$ . However,  $f'(x) = 3 - \cos x$  is never zero. So, there is no second root:  $f(x)$  has precisely one root.

**S-12:** The function  $f(x)$  is continuous and differentiable over all real numbers. If  $a$  and  $b$  are distinct roots of  $f(x)$ , then  $f'(c) = 0$  for some  $c$  strictly between  $a$  and  $b$  (Rolle's Theorem). So, let's think about  $f'(x)$ .

$$f'(x) = (4x + 1)^3 + 1$$

This is simple enough that we can find its zero explicitly:

$$(4x + 1)^3 + 1 = 0 \quad \Leftrightarrow \quad (4x + 1)^3 = -1 \quad \Leftrightarrow \quad 4x + 1 = -1 \quad \Leftrightarrow \quad 4x = -2$$

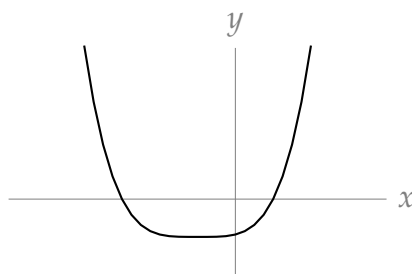
Hence  $f'(c) = 0$  only when  $c = -\frac{1}{4}$ . That the derivative only has a single zero is very useful. It means (via Rolle's theorem) that if  $f(x)$  has distinct roots  $a$  and  $b$  with  $a < b$ , then we must have  $a < -\frac{1}{4} < b$ . This also means that  $f(x)$  cannot have 3 distinct roots  $a, b$  and (say)  $q$  with  $a < b < q$ , because then Rolle's theorem would imply that  $f'(x)$  would have two zeros — one between  $a$  and  $b$  and another between  $b$  and  $q$ .

We've learned that  $f(x)$  has at most two roots, and we've learned something about where those roots can exist, if there are indeed two of them. But that means  $f(x)$  could have 0, 1, or 2 roots.

It's not easy to find a root of  $f(x)$  by inspection. But we can get a good enough picture of the graph of  $y = f(x)$  to tell exactly how many roots there are, just by exploiting the following properties of  $f(x)$  and  $f'(x)$ .

- As  $x$  tends to  $\pm\infty$ ,  $f(x)$  tends to  $+\infty$ .
- The derivative  $f'(x) = (4x + 1)^3 + 1$  is negative for  $x < -\frac{1}{4}$  and is positive for  $x > -\frac{1}{4}$ . That is,  $f(x)$  is decreasing for  $x < -\frac{1}{4}$  and increasing for  $x > -\frac{1}{4}$ .
- $f\left(-\frac{1}{4}\right) = \frac{1}{16} - \frac{1}{4} < 0$ .

This means the function must look something like the graph below:



except we are unsure of the locations of the  $x$ -intercepts. At present we only know that one is to the left of  $x = -1/2$  and one is to the right.

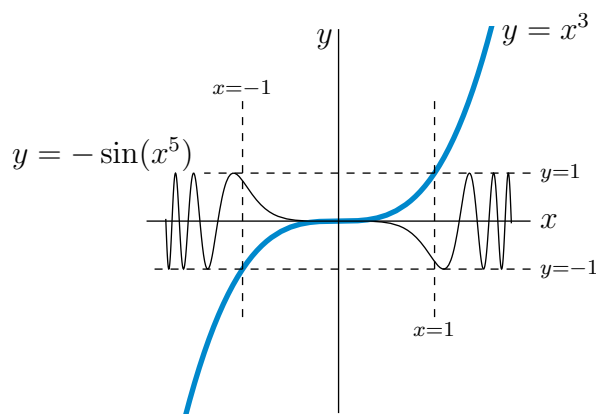
Note what happens to  $f(x)$  as  $x$  increases from strongly negative values to strongly positive values.

- When  $x$  is large and negative,  $f(x) > 0$ .
- As  $x$  increases,  $f(x)$  decreases continuously until  $x = -\frac{1}{2}$ , where  $f(x) < 0$ . In particular, since  $f(-1) = \frac{65}{16} > 0$  and  $f(-\frac{1}{2}) < 0$  and  $f(x)$  is continuous, the intermediate value theorem guarantees that  $f(x)$  takes the value zero for some  $x$  between  $-\frac{1}{2}$  and  $-1$ . More descriptively put, as  $x$  increases from hugely negative numbers to  $-\frac{1}{2}$ ,  $f(x)$  passes through zero exactly once.
- As  $x$  increases beyond  $-\frac{1}{2}$ ,  $f(x)$  increases continuously, starting negative and becoming very large and positive when  $x$  becomes large and positive. In particular, since  $f(0) = \frac{1}{16} > 0$  and  $f(-\frac{1}{2}) < 0$  and  $f(x)$  is continuous, the intermediate value theorem guarantees that  $f(x)$  takes the value zero for some  $x$  between  $-\frac{1}{2}$  and  $0$ . So, as  $x$  increases from  $-\frac{1}{2}$  to near  $+\infty$ ,  $f(x)$  again passes through zero exactly once.

So  $f(x)$  must have exactly two roots, one with  $x < -\frac{1}{2}$  and one with  $x > -\frac{1}{2}$ .

#### S-13:

- We can see by inspection that  $f(0) = 0$ , so there is at least one root. We have to determine how many any other roots there are, if any.
- The function  $f(x)$  is the sum of the two terms  $x^3$  and  $\sin(x^5)$ . A number  $x$  is a root of  $f(x)$  if and only if the two terms cancel each other exactly for that value of  $x$ . That is,  $x$  is a root of  $f$  if and only if  $x^3 = -\sin(x^5)$ . To develop some intuition in our hunt for other roots, we sketch, in the same figure, the graphs  $y = x^3$  and  $y = -\sin(x^5)$ . Then the roots of  $f(x)$  are precisely the  $x$ 's where the two curves  $y = x^3$  and  $y = -\sin(x^5)$  intersect.



- Looking at the sketch, we see that the two curves cannot possibly intersect at any point having  $|x| > 1$  — if  $|x| > 1$ , then  $|x^3| > 1$  but  $|\sin(x^5)| \leq 1$  and we cannot possibly have  $x^3 = -\sin(x^5)$ . That is, all roots of  $f(x)$  are in the interval  $[-1, 1]$ .
- From the sketch, we would probably guess that  $y = x^3$  and  $y = -\sin(x^5)$  cross only at  $x = 0$ . We can use Rolle's Theorem to verify that that is indeed the case.
- If there is a root  $a \neq 0$ , then by Rolle's Theorem (since  $f(x)$  is continuous and differentiable for all real numbers  $x$ )  $f'(c) = 0$  for some  $c$  strictly between 0 and  $a$ . In particular, since we already know any roots  $a$  will be between  $-1$  and  $1$ , if  $f(x)$  has two roots then  $f'(c) = 0$  for some  $c \in (-1, 0) \cup (0, 1)$ .
- The derivative of  $f$  is

$$f'(x) = 3x^2 + 5x^4 \cos(x^5) = x^2 (3 + 5x^2 \cos(x^5))$$

So, if  $f'(x) = 0$ , then  $x = 0$  or  $(3 + 5x^2 \cos(x^5)) = 0$ . If  $x \in (-1, 0) \cup (0, 1)$ , then

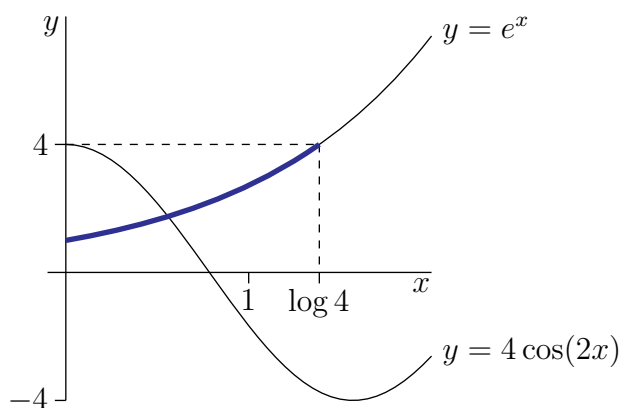
$$\begin{aligned} |x| < 1 &\implies |x^5| < 1 < \frac{\pi}{2} \implies \cos(x^5) > 0 \\ &\implies 3 + 5x^2 \cos(x^5) > 3 \\ &\implies f'(x) \neq 0 \end{aligned}$$

That is, there is no  $c \in (-1, 0) \cup (0, 1)$  with  $f'(c) = 0$ . Therefore, following our last bullet point,  $f(x)$  has only one root.

Note here that  $f'(x)$  has many zeroes — infinitely many, in fact. However,  $x = 0$  is the only root of  $f'(x)$  in the interval  $(-1, 1)$ .

#### S-14:

We are to find the number of positive solutions to the equation  $e^x = 4 \cos(2x)$ . The figure below contains the graphs of  $y = e^x$  and  $y = 4 \cos(2x)$ . The solutions to  $e^x = 4 \cos(2x)$  are precisely the  $x$ 's where  $y = e^x$  and  $y = 4 \cos(2x)$  cross.



It sure looks like there is exactly one crossing with  $x \geq 0$  and that one crossing is somewhere between  $x = 0$  and  $x = 1$ . Indeed since  $[e^x - 4 \cos(2x)]_{x=0} = -3 < 0$  and  $[e^x - 4 \cos(2x)]_{x=1} > e > 0$  and  $f(x) = e^x - 4 \cos(2x)$  is continuous, the intermediate value theorem guarantees that there is at least one root with  $0 < x < 1$ .

We still have to show that there is no second root — even if our graphs are not accurate.

Recall that the range of the cosine function is  $[-1, 1]$ . If  $e^x = 4 \cos(2x)$ , then  $e^x \leq 4$ , so  $x \leq \log(4) \approx 1.39$ . So, we only need to search for roots of  $f(x)$  on the interval  $(0, 1.4)$ : we are guaranteed there are no roots elsewhere. Over this interval,  $2x \in (0, 2.8)$ , so  $\sin(2x) > 0$ , and thus  $f'(x) = e^x + 8 \sin(2x) > 0$ . Since  $f'(x)$  has no roots in  $(0, 1.4)$ , we conclude by Rolle's Theorem that  $f(x)$  has at most one root in  $(0, 1.4)$  (and so at most one positive root total). Since we've already found that a root of  $f(x)$  exists in  $(0, 1)$ , we conclude  $e^x = 4 \cos(2x)$  has precisely one positive-valued solution.

**S-15:** (a)

$$f'(x) = 15x^4 - 30x^2 + 15 = 15(x^4 - 2x^2 + 1) = 15(x^2 - 1)^2 \geq 0$$

The derivative is nonnegative everywhere. The only values of  $x$  for which  $f'(x) = 0$  are 1 and  $-1$ , so  $f'(x) > 0$  for every  $x$  in  $(-1, 1)$ .

(b) If  $f(x)$  has two roots  $a$  and  $b$  in  $[-1, 1]$ , then by Rolle's Theorem,  $f'(c) = 0$  for some  $x$  strictly between  $a$  and  $b$ . But since  $a$  and  $b$  are in  $[-1, 1]$ , and  $c$  is between  $a$  and  $b$ , that means  $c$  is in  $(-1, 1)$ ; however, we know for every  $c$  in  $(-1, 1)$ ,  $f'(c) > 0$ , so this can't happen. Therefore,  $f(x)$  *does not* have two roots  $a$  and  $b$  in  $[-1, 1]$ . This means  $f(x)$  has at most one root in  $[-1, 1]$ .

**S-16:** Write  $f(x) = e^x$ . Since  $f(x)$  is continuous and differentiable, the Mean Value Theorem asserts that there exists some  $c$  between 0 and  $T$  such that

$$f'(c) = \frac{f(T) - f(0)}{T - 0}$$

---

The problem asks us to find this value of  $c$ . Solving:

$$\begin{aligned} e^c &= \frac{e^T - e^0}{T} \\ e^c &= \frac{e^T - 1}{T} \\ c &= \log \left( \frac{e^T - 1}{T} \right) \end{aligned}$$

**S-17:** The domains of  $\operatorname{arcsec} x$  and  $C - \operatorname{arcsec} x$  are the same:  $|x| \geq 1$ . Define  $f(x) = \operatorname{arcsec} x + \operatorname{arccsc} x$ , and note that the domain of  $f(x)$  is also  $|x| \geq 1$ . Using Theorem 2.12.7 in the CLP-1 text,

$$f'(x) = \frac{d}{dx} \operatorname{arcsec} x + \frac{d}{dx} \operatorname{arccsc} x = \frac{1}{|x|\sqrt{x^2 - 1}} + \frac{-1}{|x|\sqrt{x^2 - 1}} = 0$$

By Corollary 2.13.11 in the CLP-1 text, this means that  $f(x)$  is constant on any interval in  $|x| \geq 1$ . So  $f(x)$  is a constant, call it  $C_+$ , on  $x \geq 1$ , and  $f(x)$  is also a constant, call it  $C_-$ , on  $x \leq -1$ .

In order to find  $C_+$ , we find  $f(1)$ , because we know angles for which the secant and cosecant are  $x = 1$ .

$$\begin{aligned} \cos(0) = 1 &\implies \sec(0) = \frac{1}{1} = 1 \implies \operatorname{arcsec}(1) = 0 \\ \sin\left(\frac{\pi}{2}\right) = 1 &\implies \csc\left(\frac{\pi}{2}\right) = \frac{1}{1} = 1 \implies \operatorname{arccsc}(1) = \frac{\pi}{2} \\ &\implies C_+ = \operatorname{arcsec}(1) + \operatorname{arccsc}(1) = \frac{\pi}{2} \end{aligned}$$

In order to find  $C_-$ , we find  $f(-1)$ , because we know angles for which the secant and cosecant are  $x = -1$ .

$$\begin{aligned} \cos(\pi) = -1 &\implies \sec(\pi) = \frac{1}{-1} = -1 \implies \operatorname{arcsec}(-1) = \pi \\ \sin\left(-\frac{\pi}{2}\right) = -1 &\implies \csc\left(-\frac{\pi}{2}\right) = \frac{1}{-1} = -1 \implies \operatorname{arccsc}(-1) = -\frac{\pi}{2} \\ &\implies C_- = \operatorname{arcsec}(-1) + \operatorname{arccsc}(-1) = \frac{\pi}{2} \end{aligned}$$

So  $f(x) = \operatorname{arcsec} x + \operatorname{arccsc} x = \frac{\pi}{2}$  for all  $|x| \geq 1$  and  $\operatorname{arcsec} x = \frac{\pi}{2} - \operatorname{arccsc} x$  for all  $|x| \geq 1$ .

**S-18:** Since  $e^{-f(x)}$  is always positive (regardless of the value of  $f(x)$ ),

$$f'(x) = \frac{1}{1 + e^{-f(x)}} < \frac{1}{1 + 0} = 1$$

for every  $x$ .

Since  $f'(x)$  exists for every  $x$ , we see that  $f$  is differentiable, so the Mean Value Theorem applies. If  $f(100)$  is greater than or equal to 100, then by the Mean Value Theorem, there would have to be some  $c$  between 0 and 100 such that

$$f'(c) = \frac{f(100) - f(0)}{100} \geq \frac{100}{100} = 1$$

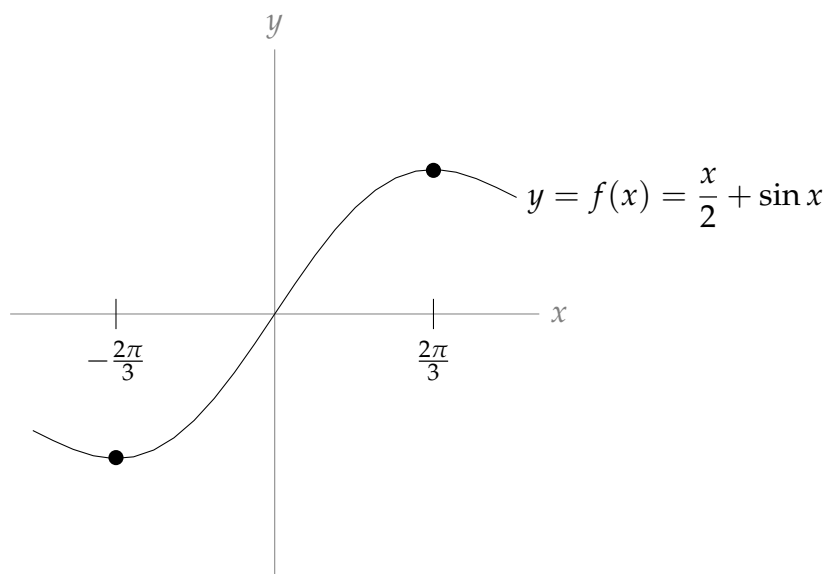
Since  $f'(x) < 1$  for every  $x$ , there is *no* value of  $c$  as described. Therefore, it is not possible that  $f(100) \geq 100$ . So,  $f(100) < 100$ .

**S-19:** If  $2x + \sin x$  is one-to-one over an interval, it never takes the same value for two distinct numbers in that interval. By Rolle's Theorem, if  $f(a) = f(b)$  for distinct  $a$  and  $b$ , then  $f'(c) = 0$  for some  $c$  between  $a$  and  $b$ . However,  $f'(x) = 2 + \cos x$ , which is never zero. In fact,  $f'(x) \geq 1$  for all  $x$ , so  $f(x)$  is strictly increasing over its entire domain. Therefore, our function  $f$  *never* takes the same value twice, so it is one-to-one over all the real numbers,  $(-\infty, \infty)$ .

When we define the inverse function  $f^{-1}(x)$ , the domain of  $f$  is the range of  $f^{-1}$ , and vice-versa. In general, we might have to *restrict* the domain of  $f$  (and hence the range of  $f^{-1}$ ) to an interval where  $f$  is one-to-one, but in our case, this isn't necessary. So, the range of  $f^{-1}$  is  $(-\infty, \infty)$  and the domain of  $f^{-1}$  is the range of  $f$ :  $(-\infty, \infty)$ .

**S-20:** If  $f(x) = \frac{x}{2} + \sin x$  is one-to-one over an interval, it never takes the same value for two distinct numbers in that interval. By Rolle's Theorem, if  $f(a) = f(b)$  for distinct  $a$  and  $b$ , then  $f'(c) = 0$  for some  $c$  between  $a$  and  $b$ . Since  $f'(x) = \frac{1}{2} + \cos x$ ,  $f'(x) = 0$  when  $x = 2n\pi \pm \frac{2\pi}{3}$  for some integer  $n$ . So, in particular, if  $a$  and  $b$  are distinct numbers in the interval  $[-\frac{2\pi}{3}, \frac{2\pi}{3}]$ , then for every  $c$  strictly between  $a$  and  $b$ ,  $f'(c) \neq 0$ , so by Rolle's Theorem  $f(a) \neq f(b)$ . Therefore  $f(x)$  is one-to-one on the interval  $[-\frac{2\pi}{3}, \frac{2\pi}{3}]$ .

We should also show that the interval  $[-\frac{2\pi}{3}, \frac{2\pi}{3}]$  cannot be extended to a larger interval over which  $f(x)$  is still one-to-one. Consider the derivative  $f'(x) = \frac{1}{2} + \cos x$ . For all  $-\frac{2\pi}{3} < x < \frac{2\pi}{3}$ , we have  $\cos x > -\frac{1}{2}$  (sketch the graph of  $\cos x$  yourself) so that  $f'(x) > 0$  and  $f(x)$  is increasing. But at  $x = \frac{2\pi}{3}$ ,  $f'(x) = 0$ , and then for  $x$  a bit bigger than  $\frac{2\pi}{3}$  we have  $\cos x < -\frac{1}{2}$  so that  $f'(x) < 0$  and  $f(x)$  is decreasing. So the graph "reverses direction", and  $f(x)$  repeats values. (See the graph of  $y = f(x)$  below.) The same is true for  $x$  a little smaller than  $-\frac{2\pi}{3}$ .



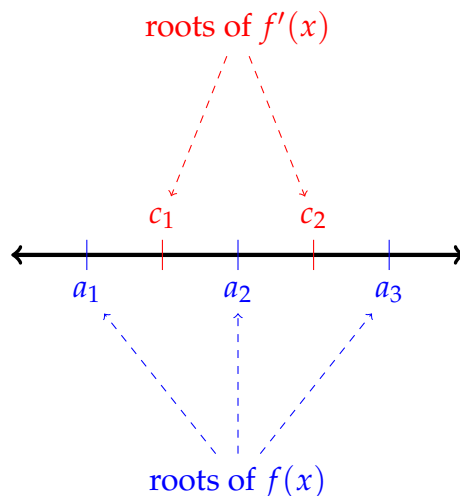
When we define the inverse function  $f^{-1}(x)$ , first we restrict  $f$  to  $[-\frac{2\pi}{3}, \frac{2\pi}{3}]$ . Then the range of  $f^{-1}$  is also  $[-\frac{2\pi}{3}, \frac{2\pi}{3}]$ . The domain of  $f^{-1}$  is the range of  $f$  over this interval, so  $[-(\frac{\pi}{3} + \frac{\sqrt{3}}{2}), (\frac{\pi}{3} + \frac{\sqrt{3}}{2})]$ .

**S-21:** Define  $h(x) = f(x) - g(x)$ , and notice  $h(a) = f(a) - g(a) < 0$  and  $h(b) = f(b) - g(b) > 0$ . Since  $h$  is the difference of two functions that are continuous over  $[a, b]$  and differentiable over  $(a, b)$ , also  $h$  is continuous over  $[a, b]$  and differentiable over  $(a, b)$ . So, by the Mean Value Theorem, there exists some  $c \in (a, b)$  with

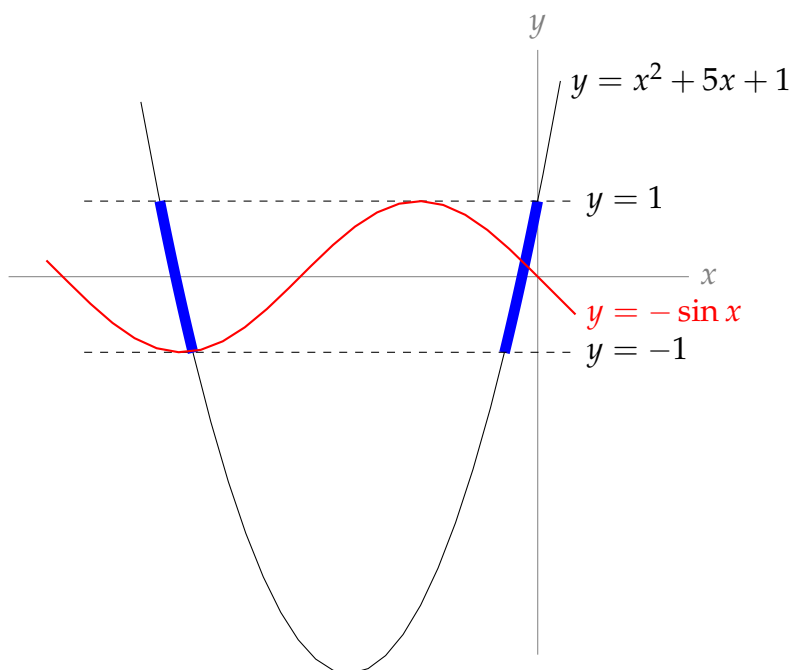
$$h'(c) = \frac{h(b) - h(a)}{b - a}$$

Since  $(a, b)$  is an interval,  $b > a$ , so the denominator of the above expression is positive; since  $h(b) > 0 > h(a)$ , also the numerator of the above expression is positive. So,  $h'(c) > 0$  for some  $c \in (a, b)$ . Since  $h'(c) = f'(c) - g'(c)$ , we conclude  $f'(c) > g'(c)$  for some  $c \in (a, b)$ .

**S-22:** Since  $f(x)$  is differentiable over all real numbers, it is also continuous over all real numbers. We claim that  $f(x)$  cannot have four or more distinct roots. For every two distinct roots  $a < b$ , Rolle's Theorem tells us there is a  $c \in (a, b)$  such that  $f'(c) = 0$ : that is,  $c$  is a root of  $f'$ . Since  $f'$  has only two distinct roots,  $f$  can have at most three distinct roots.



**S-23:** We are asked to find the number of solutions to the equation  $x^2 + 5x + 1 = -\sin x$ . The figure below contains the graphs of  $y = x^2 + 5x + 1$  and  $y = -\sin x$ . The solutions to  $x^2 + 5x + 1 = -\sin x$  are precisely the  $x$ 's where  $y = x^2 + 5x + 1$  and  $y = -\sin x$  cross.

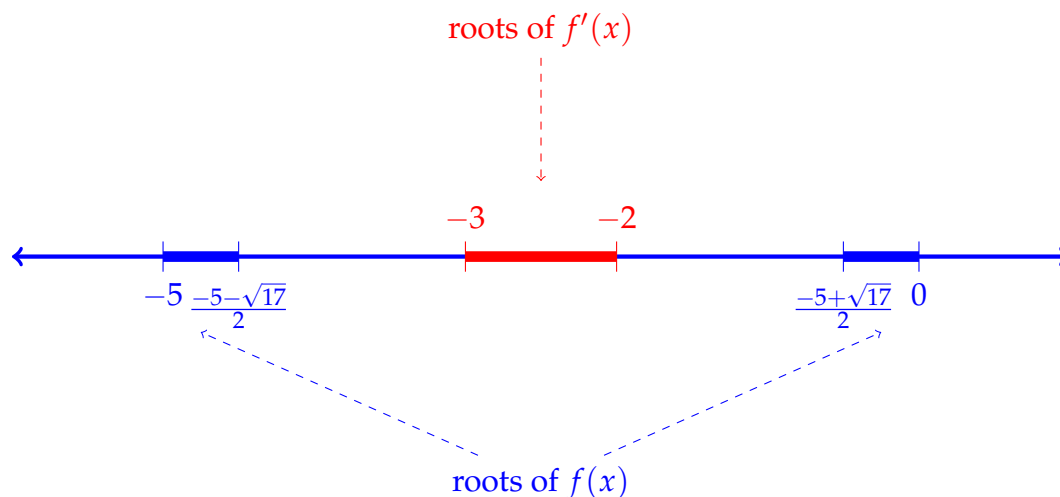


From the figure, it sure looks like there are two crossings. Since the function  $-\sin x$  has range  $[-1, 1]$ , if the two functions cross, then also  $-1 \leq x^2 + 5x + 1 \leq 1$ . This portion of the quadratic function is highlighted in blue in the figure. The  $x$  coordinates of the end points of the blue arcs are found by solving  $x^2 + 5x + 1 = \pm 1$ , i.e.  $x = 0, -5$ , and (using the quadratic equation)  $x = \frac{-5 \pm \sqrt{17}}{2}$ .

We are now in a position to exploit the intuition that we have built using the above figure to write a concise argument showing that  $f(x)$  has exactly two roots. Remember that, in general, if we want to show that a function has  $n$  roots, we need to show that there exist  $n$  distinct roots somewhere, and that there do not exist  $n + 1$  distinct roots. This argument is given below, in blue text.

- $f(x)$  is continuous over all real numbers
- $f(x) = 0$  only when  $-\sin x = x^2 + 5x + 1$ , which only happens when  $|x^2 + 5x + 1| \leq 1$ . Thus,  $f(x)$  only has roots in the intervals  $\left[-5, \frac{-5 - \sqrt{17}}{2}\right]$  and  $\left[\frac{-5 + \sqrt{17}}{2}, 0\right]$ .
- $f(-5) = \sin(-5) + 1 > 0$ , and  $f\left(\frac{-5 - \sqrt{17}}{2}\right) = \sin\left(\frac{-5 - \sqrt{17}}{2}\right) - 1 < 0$ .  
So, by the IVT,  $f(c) = 0$  for some  $c \in \left(-5, \frac{-5 - \sqrt{17}}{2}\right)$ .
- $f(0) = 1 > 0$ , and  $f\left(\frac{-5 + \sqrt{17}}{2}\right) = \sin\left(\frac{-5 + \sqrt{17}}{2}\right) - 1 < 0$ .  
So, by the IVT,  $f(c) = 0$  for some  $c \in \left(\frac{-5 + \sqrt{17}}{2}, 0\right)$ .
- $f'(x) = \cos x + 2x + 5$ . If  $f'(x) = 0$ , then  $2x + 5 = -\cos x$ , so  $|2x + 5| \leq 1$ . So, the only interval that can contain roots of  $f'(x)$  is  $[-3, -2]$ .
- Suppose  $f(x)$  has more than two roots. Then it has two roots in the interval

$\left[-5, \frac{-5-\sqrt{17}}{2}\right]$  OR it has two roots in the interval  $\left[\frac{-5+\sqrt{17}}{2}, 0\right]$ . Since  $f(x)$  is differentiable for all real numbers, Rolle's Theorem tells us that  $f'(x)$  has a root in  $\left(-5, \frac{-5-\sqrt{17}}{2}\right)$  or in  $\left(\frac{-5+\sqrt{17}}{2}, 0\right)$ . However, since all roots of  $f'(x)$  are in the interval  $[-3, -2]$ , and this interval shares no points with  $\left(-5, \frac{-5-\sqrt{17}}{2}\right)$  or  $\left(\frac{-5+\sqrt{17}}{2}, 0\right)$ , this cannot be the case. Therefore  $f(x)$  does not have more than two roots.



- Since  $f(x)$  has at least two roots, and not more than two roots,  $f(x)$  has exactly two roots.

## Solutions to Exercises [2.14](#) — Jump to [TABLE OF CONTENTS](#)

**S-1:** The derivative of  $e^x$  is  $e^x$ : taking derivatives leaves the function unchanged, even if we do it 180 times. So  $f^{(180)} = e^x$ .

**S-2:** Since  $f'(x) > 0$  over  $(a, b)$ , we know from Corollary 2.13.11 that  $f(x)$  is increasing over  $(a, b)$ , so [\(ii\)](#) holds. Since  $f''(x) > 0$ , and  $f''(x)$  is the derivative of  $f'(x)$ , by the same reasoning we see that  $f'(x)$  is increasing. Since  $f'(x)$  is the rate at which  $f(x)$  is increasing, that means that the rate at which  $f'(x)$  is increasing is itself increasing: this is, [\(iv\)](#) holds (and *not* [\(iii\)](#)).

There is no reason to think [\(i\)](#) or [\(v\)](#) holds, but to be thorough we will give an example showing that they do not need to be true. If  $f(x) = x^2 - 10$  and  $(a, b) = (0, 1)$ , then  $f'(x) = 2x > 0$  over  $(0, 1)$ , and  $f''(x) = 2 > 0$  everywhere, but  $f(x) < 0$  for all  $x \in (0, 1)$ , so [\(i\)](#) does not hold. Also,  $f'''(x) = 0$  everywhere, so [\(v\)](#) does not hold either.

**S-3:** Every time we differentiate  $f(x)$ , the constant out front gets multiplied by an ever-decreasing constant, while the power decreases by one. As in Example 2.14.2,

$$\frac{d^{15}}{dx^{15}} ax^{15} = a \cdot 15!. \text{ So, if } a \cdot 15! = 3, \text{ then } a = \frac{3}{15!}.$$

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**S-4:** The derivative  $\frac{dy}{dx}$  is  $\frac{11}{4}$  only at the point  $(1, 3)$ : it is not *constantly*  $\frac{11}{4}$ , so it is wrong to differentiate the constant  $\frac{11}{4}$  to find  $\frac{d^2y}{dx^2}$ . Below is a correct solution.

$$-28x + 2y + 2xy' + 2yy' = 0$$

Plugging in  $x = 1, y = 3$ :

$$\begin{aligned} -28 + 6 + 2y' + 6y' &= 0 \\ y' &= \frac{11}{4} \quad \text{at the point } (1, 3) \end{aligned}$$

Differentiating the equation  $-28x + 2y + 2xy' + 2yy' = 0$ :

$$\begin{aligned} -28 + 2y' + 2y' + 2xy'' + 2y'y' + 2yy'' &= 0 \\ 4y' + 2(y')^2 + 2xy'' + 2yy'' &= 28 \end{aligned}$$

At the point  $(1, 3), y' = \frac{11}{4}$ . Plugging in:

$$\begin{aligned} 4\left(\frac{11}{4}\right) + 2\left(\frac{11}{4}\right)^2 + 2(1)y'' + 2(3)y'' &= 28 \\ y'' &= \frac{15}{64} \end{aligned}$$

**S-5:**

$$\begin{aligned} f(x) &= x \log x - x \\ f'(x) &= \log x + x \cdot \frac{1}{x} - 1 \\ &= \log x \\ f''(x) &= \frac{1}{x} \end{aligned}$$

**S-6:**

$$\begin{aligned} \frac{d}{dx} \{\arctan x\} &= \frac{1}{1+x^2} \\ \frac{d}{dx} \left\{ \frac{1}{1+x^2} \right\} &= \frac{d}{dx} \left\{ (1+x^2)^{-1} \right\} \\ &= (-1)(1+x^2)^{-2}(2x) \\ &= \frac{-2x}{(1+x^2)^2} \end{aligned}$$


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**S-7:** We use implicit differentiation, twice.

$$\begin{aligned}2x + 2yy' &= 0 \\2 + (2y)y'' + (2y')y' &= 0 \\y'' &= -\frac{(y')^2 + 1}{y}\end{aligned}$$

So, we need an expression for  $y'$ . We use the equation  $2x + 2yy' = 0$  to conclude  $y' = -\frac{x}{y}$ :

$$\begin{aligned}y'' &= -\frac{\left(-\frac{x}{y}\right)^2 + 1}{y} \\&= -\frac{\frac{x^2}{y^2} + 1}{y} \\&= -\frac{x^2 + y^2}{y^3} \\&= -\frac{1}{y^3}\end{aligned}$$

**S-8:** The question asks for  $s''(1)$ . We start our differentiation using the quotient rule:

$$\begin{aligned}s'(t) &= \frac{e^t(t^2 + 1) - e^t(2t)}{(t^2 + 1)^2} \\&= \frac{e^t(t^2 - 2t + 1)}{(t^2 + 1)^2}\end{aligned}$$

Using the quotient rule again,

$$\begin{aligned}s''(t) &= \frac{(t^2 + 1)^2 \frac{d}{dt}\{e^t(t^2 - 2t + 1)\} - e^t(t^2 - 2t + 1) \frac{d}{dt}\{(t^2 + 1)^2\}}{(t^2 + 1)^4} \\&= \frac{(t^2 + 1)^2 \cdot [e^t(2t - 2) + e^t(t^2 - 2t + 1)] - e^t(t^2 - 2t + 1) \cdot 2(t^2 + 1)(2t)}{(t^2 + 1)^4} \\&= \frac{e^t(t^2 + 1)^2(t^2 - 1) - 4te^t(t - 1)^2(t^2 + 1)}{(t^2 + 1)^4} \\s''(1) &= 0\end{aligned}$$

**S-9:** We differentiate using the chain rule.

$$\frac{d}{dx}\{\log(5x^2 - 12)\} = \frac{10x}{5x^2 - 12}$$

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Using the quotient rule:

$$\begin{aligned}\frac{d^2}{dx^2} \{\log(5x^2 - 12)\} &= \frac{d}{dx} \left\{ \frac{10x}{5x^2 - 12} \right\} \\ &= \frac{(5x^2 - 12)(10) - 10x(10x)}{(5x^2 - 12)^2} \\ &= \frac{-10(5x^2 + 12)}{(5x^2 - 12)^2}\end{aligned}$$

Using the quotient rule one last time:

$$\begin{aligned}\frac{d^3}{dx^3} \{\log(5x^2 - 12)\} &= \frac{d}{dx} \left\{ \frac{-10(5x^2 + 12)}{(5x^2 - 12)^2} \right\} \\ &= \frac{(5x^2 - 12)^2(-10)(10x) + 10(5x^2 + 12)(2)(5x^2 - 12)(10x)}{(5x^2 - 12)^4} \\ &= \frac{(5x^2 - 12)(-100x) + (200x)(5x^2 + 12)}{(5x^2 - 12)^3} \\ &= \frac{100x(-5x^2 + 12 + 10x^2 + 24)}{(5x^2 - 12)^3} \\ &= \frac{100x(5x^2 + 36)}{(5x^2 - 12)^3}\end{aligned}$$

**S-10:** The velocity of the particle is given by  $h'(t) = \sin t$ . Note  $0 < 1 < \pi$ , so  $h'(1) > 0$ —the particle is rising (moving in the positive direction, in this case “up”). The acceleration of the particle is  $h''(t) = \cos t$ . Since  $0 < 1 < \frac{\pi}{2}$ ,  $h''(t) > 0$ , so  $h'(t)$  is increasing: the particle is moving up, and it’s doing so at an increasing rate. So, the particle is speeding up.

**S-11:** For this problem, remember that velocity has a sign indicating direction, while speed does not.

The velocity of the particle is given by  $h'(t) = 3t^2 - 2t - 5$ . At  $t = 1$ , the velocity of the particle is  $-4$ , so the particle is moving downwards with a speed of 4 units per second. The acceleration of the particle is  $h''(t) = 6t - 2$ , so when  $t = 1$ , the acceleration is (positive) 4 units per second per second. That means the velocity (currently  $-4$  units per second) is becoming a bigger number—since the velocity is negative, a bigger number is closer to zero, so the speed of the particle is getting smaller. (For instance, a velocity of  $-3$  represents a slower motion than a velocity of  $-4$ .) So, the particle is slowing down at  $t = 1$ .

**S-12:**

$$x^2 + x + y = \sin(xy)$$

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We differentiate implicitly. For ease of notation, we write  $y'$  for  $\frac{dy}{dx}$ .

$$2x + 1 + y' = \cos(xy)(y + xy')$$

We're interested in  $y''$ , so we implicitly differentiate again.

$$2 + y'' = -\sin(xy)(y + xy')^2 + \cos(xy)(2y' + xy'')$$

We want to know what  $y''$  is when  $x = y = 0$ . Plugging these in yields the following:

$$2 + y'' = 2y'$$

So, we need to know what  $y'$  is when  $x = y = 0$ . We can get this from the equation  $2x + 1 + y' = \cos(xy)(y + xy')$ , which becomes  $1 + y' = 0$  when  $x = y = 0$ . So, at the origin,  $y' = -1$ , and

$$\begin{aligned} 2 + y'' &= 2(-1) \\ y'' &= -4 \end{aligned}$$

Remark: a common mistake is to stop at the equation  $2x + 1 + y' = \cos(xy)(y + xy')$ , plug in  $x = y = 0$ , find  $y' = -1$ , and decide  $y'' = \frac{d}{dx}\{-1\} = 0$ . This is due to a slight sloppiness in the usual notation. When we wrote  $y' = 1$ , what we meant is that *at the point*  $(0, 0)$ ,  $\frac{dy}{dx} = -1$ . More properly written:  $\frac{dy}{dx}\bigg|_{x=0, y=0} = -1$ . This is not the same as saying  $y' = 1$  everywhere (in which case, indeed,  $y''$  would be 0 everywhere).

**S-13:** For (a) and (b), notice the following:

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x \\ \frac{d}{dx} \cos x &= -\sin x \\ \frac{d}{dx} \{-\sin x\} &= -\cos x \\ \frac{d}{dx} \{-\cos x\} &= \sin x \\ \frac{d}{dx} \sin x &= \cos x \end{aligned}$$

The fourth derivative of  $\sin x$  is  $\sin x$ , and the fourth derivative of  $\cos x$  is  $\cos x$ , so (a) and (b) are true.

$$\begin{aligned}\frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \sec^2 x &= 2 \sec x (\sec x \tan x) = 2 \sec^2 x \tan x \\ \frac{d}{dx} \{2 \sec^2 x \tan x\} &= (4 \sec x \cdot \sec x \tan x) \tan x + 2 \sec^2 x \sec^2 x \\ &= 4 \sec^2 x \tan^2 x + 2 \sec^4 x \\ \frac{d}{dx} \{4 \sec^2 x \tan^2 x + 2 \sec^4 x\} &= (8 \sec x \cdot \sec x \tan x) \tan^2 x + 4 \sec^2 x (2 \tan x \cdot \sec^2 x) \\ &\quad + 8 \sec^3 x \cdot \sec x \tan x \\ &= 8 \sec^2 x \tan^3 x + 16 \sec^4 x \tan x\end{aligned}$$

So,  $\frac{d^4}{dx^4} \tan x = 8 \sec^2 x \tan^3 x + 16 \sec^4 x \tan x$ . It certainly seems like this is not the same as  $\tan x$ , but remember that sometimes trig identities can fool you:  $\tan^2 x + 1 = \sec^2 x$ , and so on. So, to be absolutely sure that these are not equal, we need to find a value of  $x$  so that the output of one is not the same as the output of the other. When  $x = \frac{\pi}{4}$ :

$$8 \sec^2 x \tan^3 x + 16 \sec^4 x \tan x = 8 (\sqrt{2})^2 (1)^3 + 16 (\sqrt{2})^4 (1) = 80 \neq 1 = \tan x.$$

So, (c) is false.

**S-14:** Since  $f'(x) < 0$ , we need a decreasing function. This only applies to (ii), (iii), and (v). Since  $f''(x) > 0$ , that means  $f'(x)$  is increasing, so the *slope* of the function must be increasing. In (v), the slope is constant, so  $f''(x) = 0$ —therefore, it's not (v). In (iii), the slope is decreasing, because near  $a$  the curve is quite flat ( $f'(x)$  near zero) but near  $b$  the curve is very steeply decreasing ( $f'(x)$  is a large *negative* number), so (iii) has a negative second derivative. By contrast, in (ii), the line starts out as steeply decreasing ( $f'(x)$  is a strongly negative number) and becomes flatter and flatter ( $f'(x)$  nears 0), so  $f'(x)$  is increasing—in other words,  $f''(x) > 0$ . So, (ii) is the only curve that has  $f'(x) < 0$  and  $f''(x) > 0$ .

**S-15:** We differentiate a few times to find the pattern.

$$\begin{aligned}\frac{d}{dx} \{2^x\} &= 2^x \log 2 \\ \frac{d^2}{dx^2} \{2^x\} &= 2^x \log 2 \cdot \log 2 = 2^x (\log 2)^2 \\ \frac{d^3}{dx^3} \{2^x\} &= 2^x (\log 2)^2 \cdot \log 2 = 2^x (\log 2)^3\end{aligned}$$

Every time we differentiate, we multiply the original function by another factor of  $\log 2$ . So, the  $n$ th derivative is given by:

$$\frac{d^n}{dx^n} \{2^x\} = 2^x (\log 2)^n$$

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**S-16:** We differentiate using the power rule.

$$\begin{aligned}\frac{df}{dx} &= 3ax^2 + 2bx + c \\ \frac{d^2f}{dx^2} &= 6ax + 2b \\ \frac{d^3f}{dx^3} &= 6a \\ \frac{d^4f}{dx^4} &= 0\end{aligned}$$

In the above work, remember that  $a$ ,  $b$ ,  $c$ , and  $d$  are all constants. Since they are nonzero constants,  $\frac{d^3f}{dx^3} = 6a \neq 0$ . So, the fourth derivative is the first derivative to be identically zero:  $n = 4$ .

**S-17:** (a) Using the chain rule for  $f(x)$ :

$$\begin{aligned}f'(x) &= (1 + 2x)e^{x+x^2} \\ f''(x) &= (1 + 2x)(1 + 2x)e^{x+x^2} + (2)e^{x+x^2} = (4x^2 + 4x + 3)e^{x+x^2} \\ h'(x) &= 1 + 3x \\ h''(x) &= 3\end{aligned}$$

(b)  $f(0) = h(0) = 1$ ;  $f'(0) = h'(0) = 1$ ;  $f''(0) = h''(0) = 3$

(c)  $f$  and  $h$  "start at the same place," since  $f(0) = h(0)$ . If it were clear that  $f'(x)$  were greater than  $h'(x)$  for  $x > 0$ , then we would know that  $f$  grows faster than  $h$ , so we could conclude that  $f(x) > h(x)$ , as desired. Unfortunately, it is not obvious whether  $(1 + 2x)e^{x+x^2}$  is always greater than  $1 + 3x$  for positive  $x$ . So, we look to the second derivative.  $f'(0) = h'(0)$ , and  $f''(x) = (4x^2 + 4x + 3)e^{x+x^2} > 3e^{x+x^2} > 3 = h''(x)$  when  $x > 0$ . Since  $f'(0) = h'(0)$ , and since  $f'$  grows faster than  $h'$  for positive  $x$ , we conclude  $f'(x) > h'(x)$  for all positive  $x$ . Now we can conclude that (since  $f(0) = h(0)$  and  $f$  grows faster than  $h$  when  $x > 0$ ) also  $f(x) > h(x)$  for all positive  $x$ .

**S-18:**

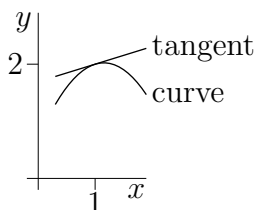
(a) We differentiate implicitly.

$$\begin{aligned}x^3y(x) + y(x)^3 &= 10x \\ 3x^2y(x) + x^3y'(x) + 3y(x)^2y'(x) &= 10\end{aligned}$$

Subbing in  $x = 1$  and  $y(1) = 2$  gives

$$\begin{aligned}(3)(1)(2) + (1)y'(1) + (3)(4)y'(1) &= 10 \\ 13y'(1) &= 4 \\ y'(1) &= \frac{4}{13}\end{aligned}$$

- (b) From part (a), the slope of the curve at  $x = 1$ ,  $y = 2$  is  $\frac{4}{13}$ , so the curve is increasing, but fairly slowly. The angle of the tangent line is  $\tan^{-1}\left(\frac{4}{13}\right) \approx 17^\circ$ . We are also told that  $y''(1) < 0$ . So the slope of the curve is decreasing as  $x$  passes through 1. That is, the line is more steeply increasing to the left of  $x = 1$ , and its slope is decreasing (getting less steep, then possibly the slope even becomes negative) as we move past  $x = 1$ .



**S-19:** (a) Using the product rule,

$$g''(x) = [f'(x) + f''(x)]e^x + [f(x) + f'(x)]e^x = [f(x) + 2f'(x) + f''(x)]e^x$$

(b) Using the product rule and our answer from (a),

$$\begin{aligned} g'''(x) &= [f'(x) + 2f''(x) + f'''(x)]e^x + [f(x) + 2f'(x) + f''(x)]e^x \\ &= [f(x) + 3f'(x) + 3f''(x) + f'''(x)]e^x \end{aligned}$$

(c) We notice that the coefficients of the derivatives of  $f$  correspond to the entries in the rows of Pascal's Triangle.

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

Pascal's Triangle

- In the first derivative of  $g$ , the coefficients of  $f$  and  $f'$  correspond to the entries in the second row of Pascal's Triangle.
- In the second derivative of  $g$ , the coefficients of  $f$ ,  $f'$ , and  $f''$  correspond to the entries in the third row of Pascal's Triangle.
- In the third derivative of  $g$ , the coefficients of  $f$ ,  $f'$ ,  $f''$ , and  $f'''$  correspond to the entries in the fourth row of Pascal's Triangle.
- We guess that, in the fourth derivative of  $g$ , the coefficients of  $f$ ,  $f'$ ,  $f''$ ,  $f'''$ , and  $f^{(4)}$  will correspond to the entries in the fifth row of Pascal's Triangle.

That is, we guess

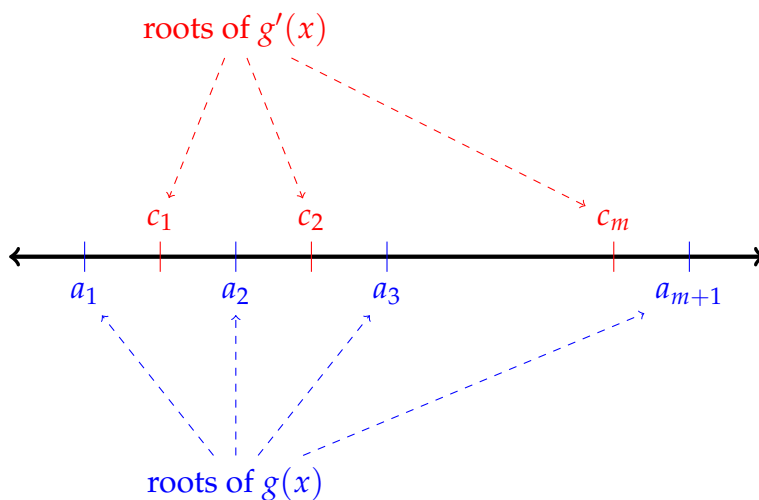
$$g^{(4)}(x) = [f(x) + 4f'(x) + 6f''(x) + 4f'''(x) + f^{(4)}(x)]e^x$$

This is verified by differentiating our answer from (a) using the product rule:

$$\begin{aligned} g'''(x) &= [f(x) + 3f'(x) + 3f''(x) + f'''(x)]e^x \\ g^{(4)}(x) &= [f'(x) + 3f''(x) + 3f'''(x) + f^{(4)}(x)]e^x + [f(x) + 3f'(x) + 3f''(x) + f'''(x)]e^x \\ &= [f(x) + 4f'(x) + 6f''(x) + 4f'''(x) + f^{(4)}(x)]e^x. \end{aligned}$$

**S-20:** Since  $f(x)$  is differentiable over all real numbers, it is also continuous over all real numbers. Similarly,  $f'(x)$  is differentiable over all real numbers, so it is also continuous over all real numbers, and so on for the first  $n$  derivatives of  $f(x)$ .

Rolle's Theorem tells us that if  $a$  and  $b$  are distinct roots of a function  $g$ , then  $g'(x) = 0$  for some  $c$  in  $(a, b)$ . That is,  $g'$  has a root strictly between  $a$  and  $b$ . Expanding this idea, if  $g$  has  $m + 1$  distinct roots, then  $g'$  must have at least  $m$  distinct roots, as in the sketch below.



So, if  $f^{(n)}(x)$  has only  $m$  roots, then  $f^{(n-1)}(x)$  has at most  $m + 1$  roots. Similarly, since  $f^{(n-1)}(x)$  has at most  $m + 1$  roots,  $f^{(n-2)}(x)$  has at most  $m + 2$  roots. Continuing in this way, we see  $f(x) = f^{(n-n)}(x)$  has at most  $m + n$  distinct roots.

**S-21:**

- Let's begin by noticing that the domain of  $f(x)$  is  $(-1, \infty)$ .
- By inspection,  $f(0) = 0$ , so  $f(x)$  has at least one root.
- If  $x \in (-1, 0)$ , then  $(x + 1)$  is positive,  $\log(x + 1)$  is negative,  $\sin(x)$  is negative, and  $-x^2$  is negative. Therefore, if  $x < 0$  is in the domain of  $f$ , then  $f(x) < 0$ . So,  $f(x)$  has no negative roots. We focus our attention on the case  $x > 0$ .
- $f'(x) = 1 - 2x + \log(x + 1) + \cos x$ . We would like to know how many positive roots  $f'(x)$  has, but it isn't obvious. So, let's differentiate again.
- $f''(x) = -2 + \frac{1}{x+1} - \sin x$ . When  $x > 0$ ,  $\frac{1}{x+1} < 1$ , so  $f''(x) < -1 - \sin(x) \leq 0$ , so  $f''(x)$  has no positive roots. Since  $f'(x)$  is continuous and differentiable over  $(0, \infty)$ ,

and since  $f''(x) \neq 0$  for all  $x \in (0, \infty)$ , by Rolle's Theorem,  $f'(x)$  has at most one root in  $[0, \infty)$ .

- Since  $f(x)$  is continuous and differentiable over  $[0, \infty)$ , and  $f'(x)$  has at most one root in  $(0, \infty)$ , by Rolle's Theorem  $f(x)$  has at most two distinct roots in  $[0, \infty)$ . (Otherwise,  $f(a) = f(b) = f(c) = 0$  for some values  $0 \leq a < b < c$ , so  $f'(d) = f'(e) = 0$  for some  $d \in (a, b)$  and some  $e \in (b, c)$ , but since  $f'(x)$  has at most one root, this is impossible.)
- We know  $f(0) = 0$ , so the remaining question is whether or not  $f(x)$  has a second root (which would have to be positive). As usual, we can show another root exists using the intermediate value theorem. We see that for large values of  $x$ ,  $f(x)$  is negative, for example:

$$f(4) = 5 \log 5 + \sin(4) - (4)^2 < 5 \log(e^2) + 1 - 16 = 11 - 16 < 0$$

For positive values of  $x$  closer to zero, we hope to find a positive value of  $f(x)$ . However, it's quite difficult to get a number  $c$  that obviously gives  $f(c) > 0$ . It suffices to observe that  $f(0) = 0$  and  $f'(0) = 2 > 0$ . From the definition of the derivative, we can conclude  $f(x) > 0$  for some  $x > 0$ . (If it is not true that  $f(x) > 0$  for some  $x > 0$ , then  $f(x) \leq 0$  for all  $x > 0$ . The definition of the derivative tells us that [since  $f'(0)$  exists]  $f'(0) = \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(h)}{h}$ ; the denominator is positive, so if the numerator were always less than or equal to zero, the limit would be less than or equal to zero as well. However, the derivative is positive, so  $f(x) > 0$  for some  $x > 0$ .) Therefore,  $f(x)$  has a second root, so  $f(x)$  has precisely two roots.

**S-22:** (a) In order to make  $f(x)$  a little more tractable, let's change the format. Since

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}, \text{ then:}$$

$$f(x) = \begin{cases} -x^2 & x < 0 \\ x^2 & x \geq 0. \end{cases}$$

Now, we turn to the definition of the derivative to figure out whether  $f'(0)$  exists.

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - 0}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} \quad \text{if it exists.}$$

Since  $f$  looks different to the left and right of 0, in order to evaluate this limit, we look at the corresponding one-sided limits. Note that when  $h$  approaches 0 from the right,  $h > 0$  so  $f(h) = h^2$ . By contrast, when  $h$  approaches 0 from the left,  $h < 0$  so  $f(h) = -h^2$ .

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(h)}{h} &= \lim_{h \rightarrow 0^+} \frac{h^2}{h} = \lim_{h \rightarrow 0^+} h = 0 \\ \lim_{h \rightarrow 0^-} \frac{f(h)}{h} &= \lim_{h \rightarrow 0^-} \frac{-h^2}{h} = \lim_{h \rightarrow 0^-} -h = 0 \end{aligned}$$

Since both one-sided limits exist and are equal to 0,

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 0$$